

ADVANCED NETWORK REPORT

Title Page

- **Project Title: Midterm Project**
- **Course Title: Networking Fundamentals II**
- **Student's Name: Owen Phillips**

CONTENTS

1. Network Topology	2
2. Network Design and Subnetting Plan	3
3. Configuration Steps and Commands	3
3.1. Basic Network Setup	4
3.2. DHCP and IPv4/IPv6 Addressing.....	8
3.3. VLAN and Trunking.....	15
3.4. FHRP Configuration.....	18
3.5. Routing Configuration	22
3.6. Other Server Configuration	25
3.7. Access Control Lists.....	38
3.8. Wireless Network and RADIUS Setup	39
4. Testing and Verification	47
5. Challenges and Solutions	60
6. Conclusion	61

1. Network Topology

Include the network **topology from Packet Tracer here**, ensuring the following are labeled:

- HQ, Branch Office 1, Branch Office 2
- All routers, switches, servers, wireless devices, and PCs/laptops
- VLANs and IP addressing schemes

2. Network Design and Subnetting Plan

Subnetting Table:

Department	Subnet	Default Gateway	Hosts Supported	First Usable IP	Last Usable IP
IT Department	10.0.0.0/25	10.0.0.126	126	192.168.1.1	192.168.1.126
Marketing Department	10.0.0.128/26 2001:DB8:ACAD:2::/64	10.0.0.132 2001:DB8:ACAD:2::4	62	10.0.0.129	10.0.0.192
Finance Department	10.0.0.192/26	10.0.0.254	62	10.0.0.193	10.0.0.254
Sales Department	10.0.1.0/26 2001:DB8:ACAD:1::/64	10.0.1.4 2001:DB8:ACAD:1::4	62	10.0.1.1	10.0.1.62
HR Department	10.0.1.64/27 2001:DB8:ACAD:3::/64	10.0.1.93 2001:DB8:ACAD:3::4	30	10.0.1.65	10.0.1.94
Support Department	10.0.1.96/27	10.0.1.100	30	10.0.1.97	10.0.1.126
Customer Service Department	10.0.1.128/27	10.0.1.132	30	10.0.1.129	10.0.1.158
Logistics Department	10.0.1.160/27	10.0.1.164	30	10.0.1.161	10.0.1.190
Research Department	10.0.1.192/27	10.0.1.193	30	10.0.1.193	10.0.1.222
Server Subnet	10.0.1.224/28	10.0.1.238	14	10.0.1.225	10.0.1.238
WLC VLAN	10.0.1.240/29	10.0.1.246	6	10.0.1.241	10.0.1.246
WAN Link 1	198.51.100.0/30 2001:DB8:ACAD:5::/64		2	198.51.100.1	198.51.100.2
WAN Link 2	198.51.100.4/30 2001:DB8:ACAD:6::/64		2	198.51.100.5	198.51.100.6
WAN Link 3	198.51.100.8/30 2001:DB8:ACAD:7::/64		2	198.51.100.9	198.51.100.10
WAN Link 4	198.51.100.12/30 2001:DB8:ACAD:8::/64		2	198.51.100.13	198.51.100.14

3. Configuration Steps and Commands

3.1. Basic Network Setup

Configuration of switches, routers, PCs, and connections

Switches:

- Set VLANs on all of them
- WHY? increase security in each department
- HOW?
- Switch(config)# vlan <vlan-id>
- Switch(config-vlan)#name <vlan-name>

```
SW1(config)#vlan 10
SW1(config-vlan)#name IT-Department
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name Finance-Department
SW1(config-vlan)#exit
SW1(config)#vlan 30
SW1(config-vlan)#name HR-Department
SW1(config-vlan)#exit
SW1(config)#vlan 40
SW1(config-vlan)#name Server-VLAN
SW1(config-vlan)#exit
SW1(config)#vlan 50
SW1(config-vlan)#name HQ-Management-VLAN
SW1(config-vlan)#exit
SW1(config)#vlan 60
SW1(config-vlan)#name Sales-Department
SW1(config-vlan)#exit
SW1(config)#vlan 70
SW1(config-vlan)#name Support-Department
SW1(config-vlan)#exit
SW1(config)#vlan 80
SW1(config-vlan)#name Logistics-Department
SW1(config-vlan)#exit
SW1(config)#vlan 90
SW1(config-vlan)#name Research-Department
VLAN #100 and #90 have an identical name: Research-Department
SW1(config-vlan)#name Marketing-Department
SW1(config-vlan)#exit
SW1(config)#vlan 100
SW1(config-vlan)#name Research-Department
SW1(config-vlan)#exit
SW1(config)#vlan 110
SW1(config-vlan)#name Customer-Service-Department
SW1(config-vlan)#exit
SW1(config)#
```

Fig 1: Example on how to set vlans on SW1

Routers:

- No shutdown on the ports being used
- WHY? So that the router can communicate
- Set up IPv6 unicast routing
- WHY? So that routers can route IPv6
- HOW?
- Router(config)#ipv6 unicast-routing
- Router(config)# int <interface>
- Router(config-if)# no shut

```
R3(config)#int g0/0
R3(config-if)#no shut
```

```
R3(config-if)#
```

```
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.60, changed state to down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.60, changed state to up
```

```
exit
```

```
R3(config)#
```

Fig 2: Example of using no shutdown on R3

PC's:

- Connect PC's to switches within the VLAN
- WHY? Increased protection
- HOW?

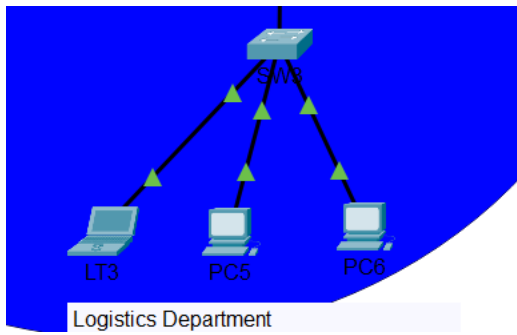
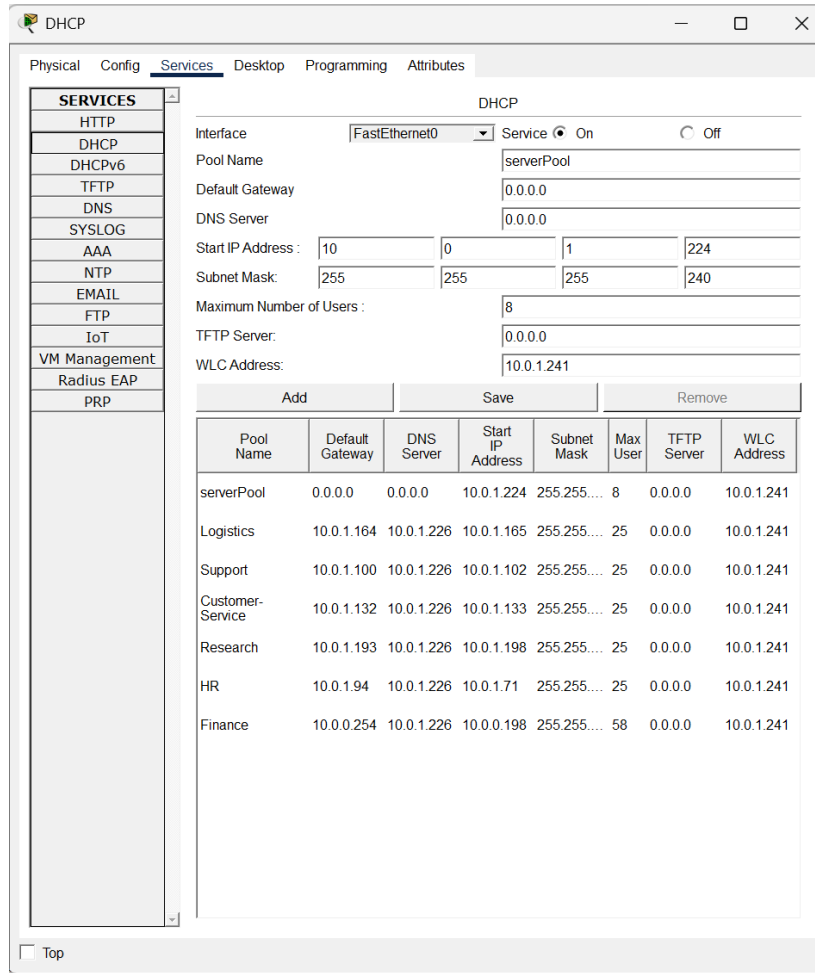


Fig 3: Example of Logistics PCs connected to their switch with VLAN

3.2. DHCP and IPv4/IPv6 Addressing

DHCP:

- Each department configured. Excluding the first 5 addresses in each subnet
- WHY? So, they can be provided with all the IP information instead of manually configuring. Also increased security
- HOW?



The screenshot shows the DHCP configuration window with the 'Services' tab selected. The 'DHCP' service is enabled for the 'FastEthernet0' interface. The configuration fields are as follows:

- Interface: FastEthernet0
- Service: On
- Pool Name: serverPool
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0
- Start IP Address: 10.0.1.224
- Subnet Mask: 255.255.255.240
- Maximum Number of Users: 8
- TFTP Server: 0.0.0.0
- WLC Address: 10.0.1.241

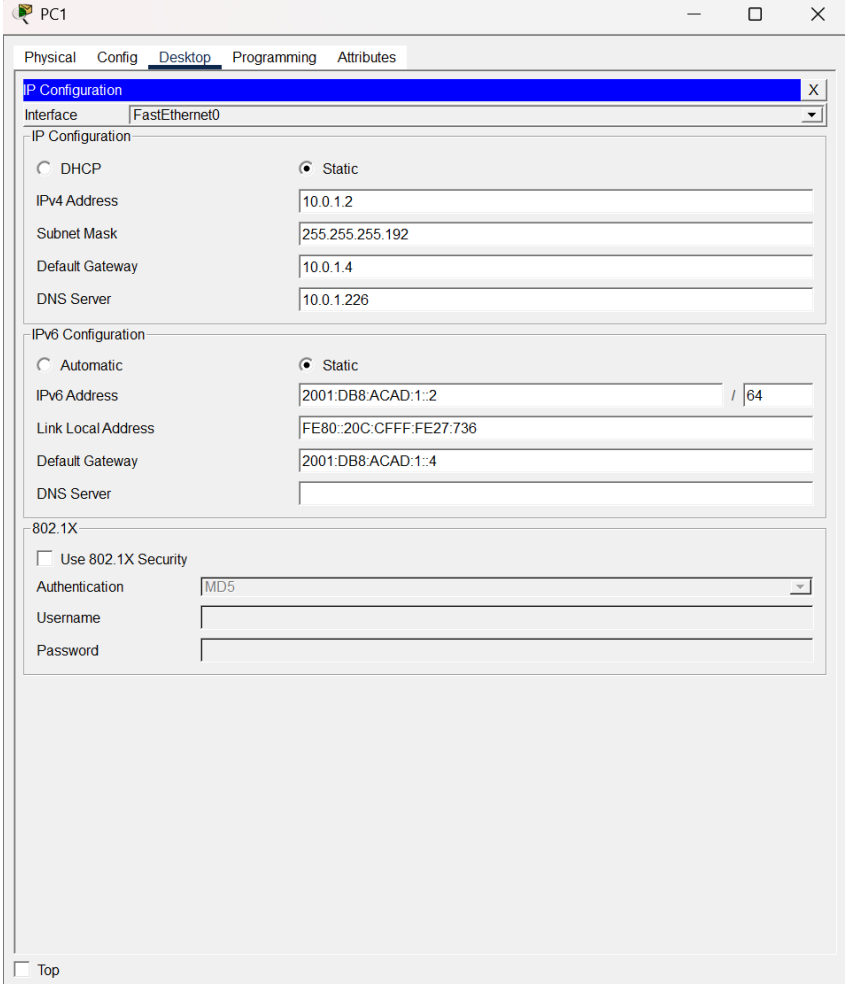
Below the configuration fields is a table of DHCP pools:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	10.0.1.224	255.255.255.240	8	0.0.0.0	10.0.1.241
Logistics	10.0.1.164	10.0.1.226	10.0.1.165	255.255.255.250	25	0.0.0.0	10.0.1.241
Support	10.0.1.100	10.0.1.226	10.0.1.102	255.255.255.250	25	0.0.0.0	10.0.1.241
Customer-Service	10.0.1.132	10.0.1.226	10.0.1.133	255.255.255.250	25	0.0.0.0	10.0.1.241
Research	10.0.1.193	10.0.1.226	10.0.1.198	255.255.255.250	25	0.0.0.0	10.0.1.241
HR	10.0.1.194	10.0.1.226	10.0.1.171	255.255.255.250	25	0.0.0.0	10.0.1.241
Finance	10.0.0.254	10.0.1.226	10.0.0.198	255.255.255.250	58	0.0.0.0	10.0.1.241

Fig 4: Example of DHCP Pools Configuration

IP Configurations

- Departments like IT have static addresses so they type in their IP address, Subnet, Default Gateway, and DNS Server
- WHY? They don't have a DHCP pool
- HOW?



The screenshot shows a network simulator window titled "PC1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Config" tab is active, and the "IP Configuration" section is expanded. The interface is set to "FastEthernet0".

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.0.1.2

Subnet Mask: 255.255.255.192

Default Gateway: 10.0.1.4

DNS Server: 10.0.1.226

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: 2001:DB8:ACAD:1::2 / 64

Link Local Address: FE80::20C:CFFF:FE27:736

Default Gateway: 2001:DB8:ACAD:1::4

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Fig 5: Static PC Configuration in Sales

- Departments like Finance have DHCP to receive their IPv4 automatically
- HOW?

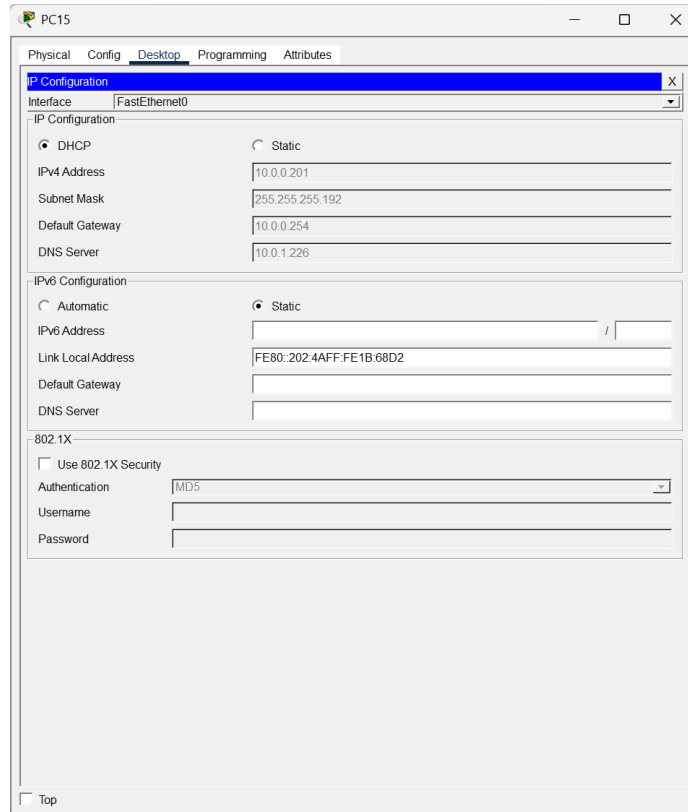


Fig 6: DHCP PC Configuration in Finance

- IP Addresses on Routers added accordingly
- WHY? If not added correctly routing could not work and they can't communicate
- HOW?
- Router(config-if)#ip address <ip-address> <subnet-mask>
- Router(config-if)#ipv6 address <ipv6-address>

```
!  
interface Serial0/0/0  
  ip address 198.51.100.6 255.255.255.252  
  ipv6 address 2001:DB8:ACAD:6::2/64  
  clock rate 2000000  
!  
interface Serial0/0/1  
  ip address 198.51.100.2 255.255.255.252  
  ipv6 address 2001:DB8:ACAD:5::2/64  
  clock rate 2000000  
!  
interface Serial0/1/0  
  ip address 198.51.100.14 255.255.255.252  
  ipv6 address 2001:DB8:ACAD:8::2/64  
!  
interface Serial0/1/1  
  ip address 198.51.100.10 255.255.255.252  
  ipv6 address 2001:DB8:ACAD:7::2/64  
!
```

Fig 7: Example of Router IP Configuration with R3

- DHCP Server needs a static IP Address, Subnet, Default Gateway, and DNS Server
- WHY? So other subnets can get their information from there. Having multiple addresses for a DHCP server won't work
- HOW?

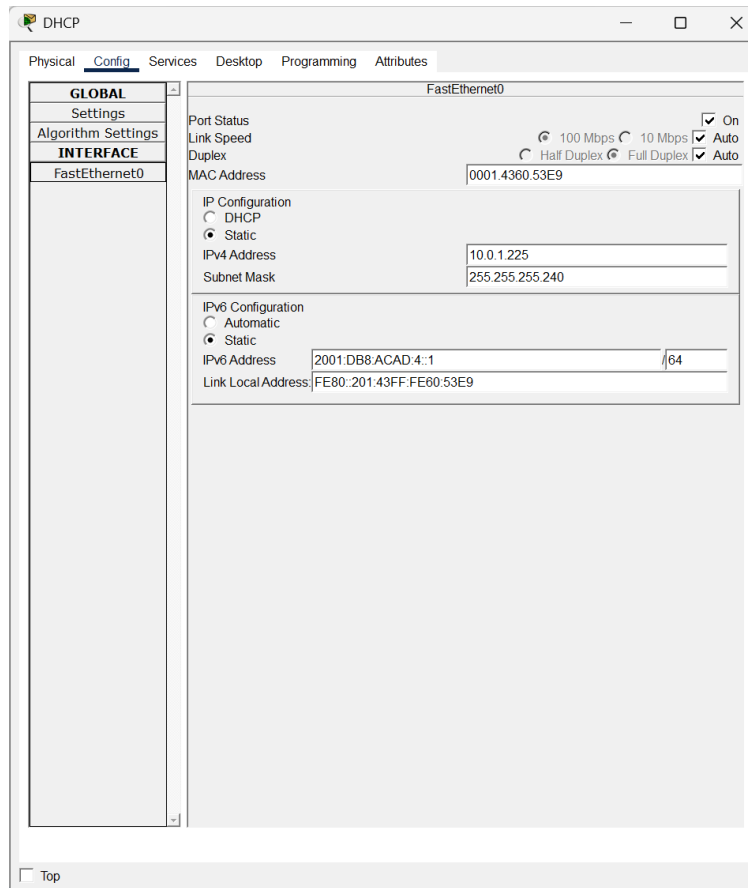
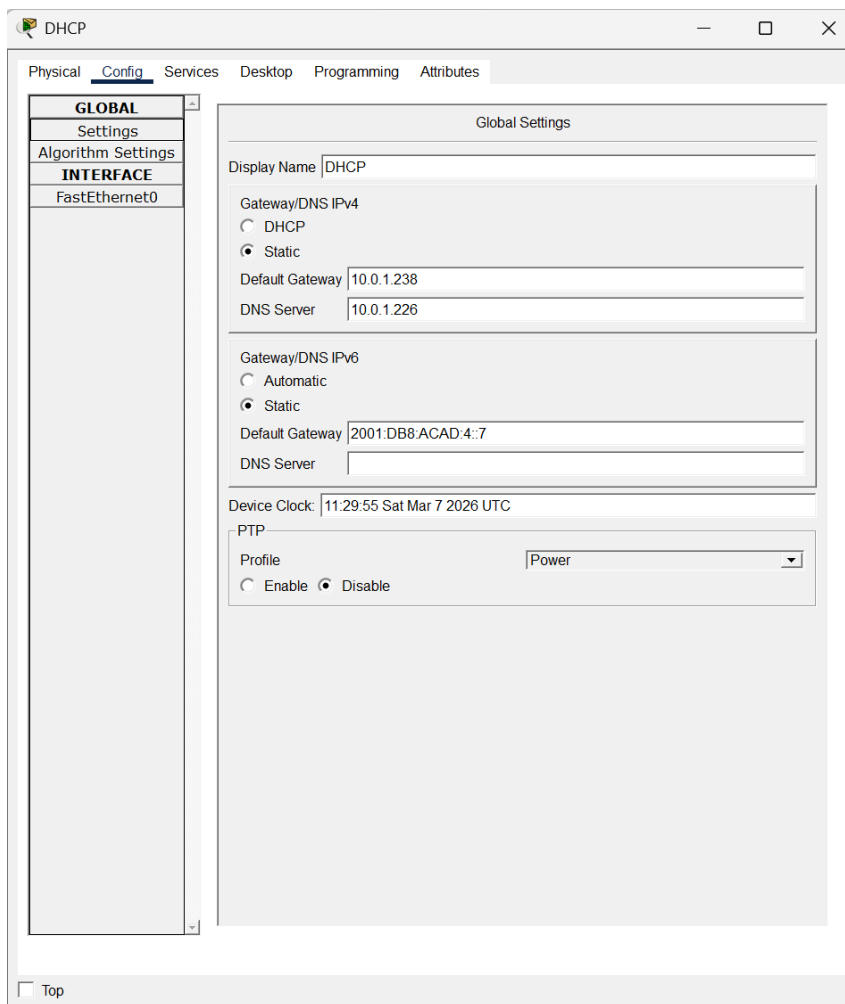


Fig 8: Example of DHCP Server Configuration



The screenshot shows a web-based configuration interface for a DHCP server. The window title is "DHCP". The left sidebar contains a tree view with the following structure:

- GLOBAL**
 - Settings
 - Algorithm Settings
- INTERFACE**
 - FastEthernet0

The main content area is titled "Global Settings" and contains the following configuration fields:

- Display Name:** DHCP
- Gateway/DNS IPv4:**
 - ☐ DHCP
 - ☒ Static
 - Default Gateway:** 10.0.1.238
 - DNS Server:** 10.0.1.226
- Gateway/DNS IPv6:**
 - ☐ Automatic
 - ☒ Static
 - Default Gateway:** 2001:DB8:ACAD:4::7
 - DNS Server:** (empty field)
- Device Clock:** 11:29:55 Sat Mar 7 2026 UTC
- PTP:**
 - Profile:** Power (dropdown menu)
 - ☐ Enable ☒ Disable

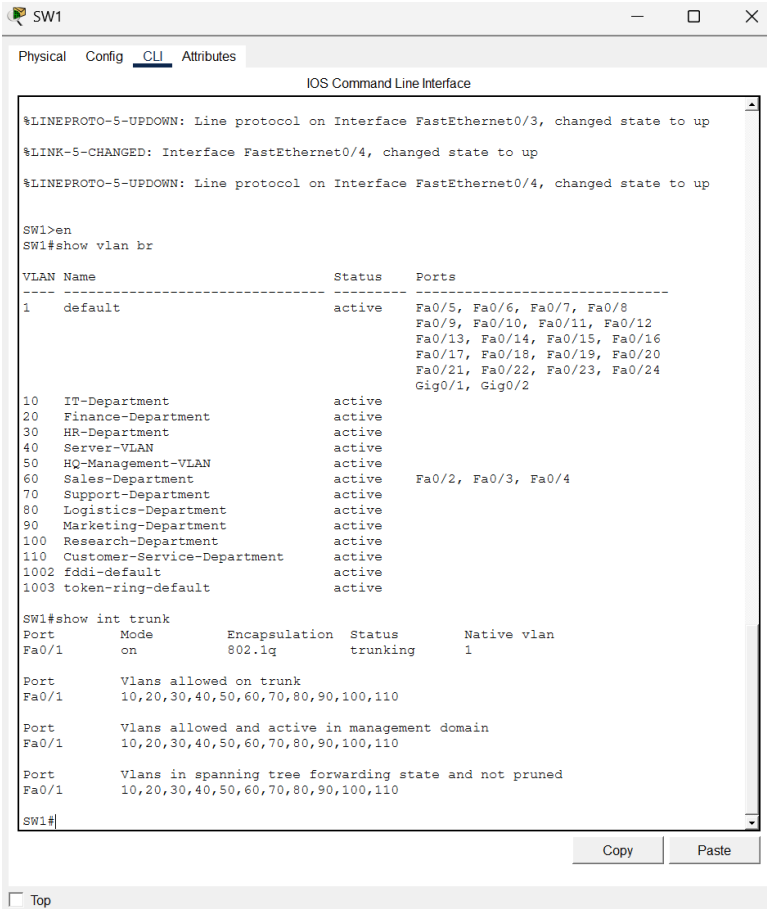
At the bottom left of the window, there is a "Top" link.

Fig 9: Example of DHCP Server Configuration

3.3. VLAN and Trunking

VLAN

- Set up switchport access VLAN on ports of PC's in vlan
- WHY? More security and to make communication between the PC's easier
- HOW?
- Switch(config-if)#switchport mode access
- Switch(config-if)#switchport access vlan <vlan-id>



```

SW1
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

SW1>en
SW1#show vlan br

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   IT-Department            active
20   Finance-Department      active
30   HR-Department            active
40   Server-VLAN              active
50   HQ-Management-VLAN       active
60   Sales-Department         active    Fa0/2, Fa0/3, Fa0/4
70   Support-Department       active
80   Logistics-Department     active
90   Marketing-Department     active
100  Research-Department      active
110  Customer-Service-Department active
1002 fddi-default            active
1003 token-ring-default     active

SW1#show int trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50,60,70,80,90,100,110

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50,60,70,80,90,100,110

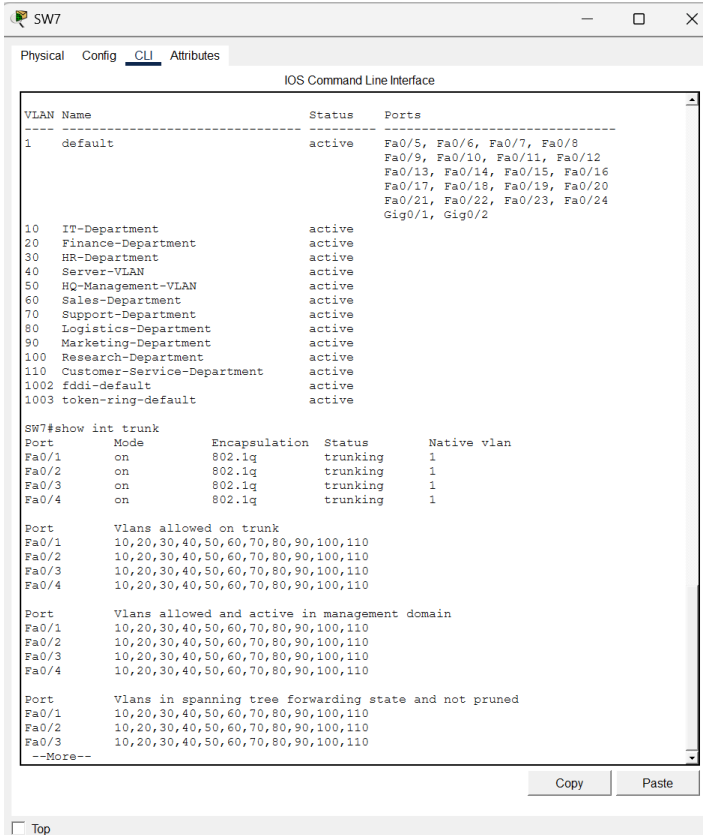
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50,60,70,80,90,100,110

SW1#
  
```

Fig 10: VLANs and Trunk of SW1

Trunk

- Set up switchport trunk on port that leads to the router interface
- WHY? To allow all the VLAN traffic to go through the trunk
- HOW?
- Switch(config-if)#switchport mode trunk
- Switch(config-if)#switchport trunk allowed vlan <vlan-id>



```

SW7
-----
Physical Config CLI Attributes
IOS Command Line Interface

VLAN Name                Status    Ports
-----
1  default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                   Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                   Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                   Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                   Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                   Gig0/1, Gig0/2

10  IT-Department          active
20  Finance-Department     active
30  HR-Department          active
40  Server-VLAN            active
50  HQ-Management-VLAN     active
60  Sales-Department       active
70  Support-Department     active
80  Logistics-Department   active
90  Marketing-Department   active
100 Research-Department   active
110 Customer-Service-Department active
1002 fddi-default         active
1003 token-ring-default   active

SW7#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50,60,70,80,90,100,110
Fa0/2     10,20,30,40,50,60,70,80,90,100,110
Fa0/3     10,20,30,40,50,60,70,80,90,100,110
Fa0/4     10,20,30,40,50,60,70,80,90,100,110

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50,60,70,80,90,100,110
Fa0/2     10,20,30,40,50,60,70,80,90,100,110
Fa0/3     10,20,30,40,50,60,70,80,90,100,110
Fa0/4     10,20,30,40,50,60,70,80,90,100,110

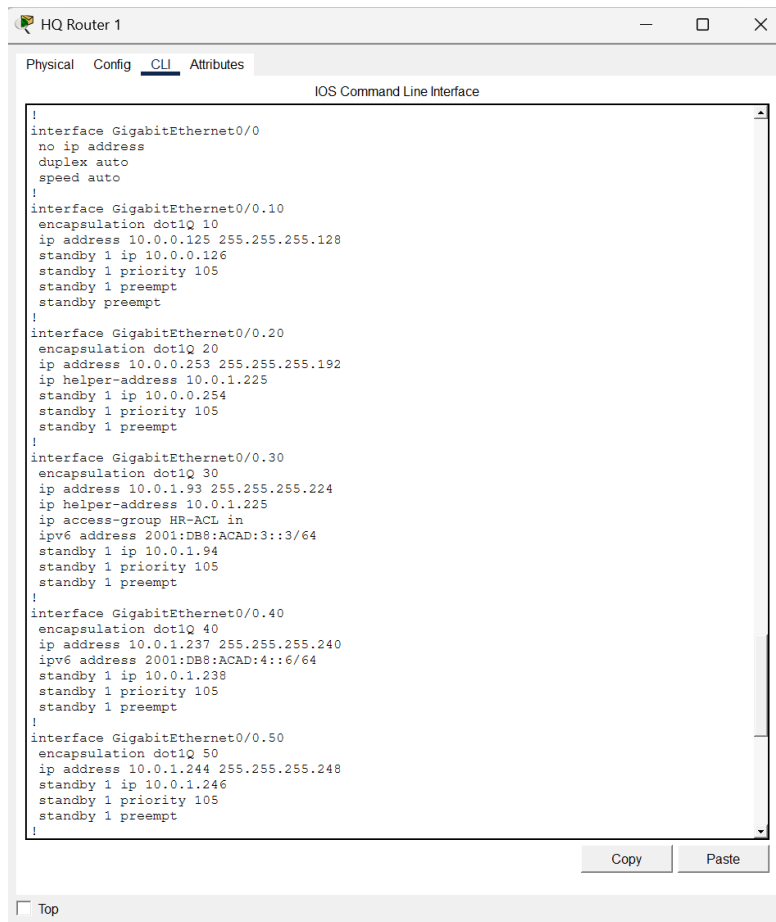
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50,60,70,80,90,100,110
Fa0/2     10,20,30,40,50,60,70,80,90,100,110
Fa0/3     10,20,30,40,50,60,70,80,90,100,110
--More--
  
```

Fig 11: VLANs and Trunk of SW7

3.4. FHRP Configuration

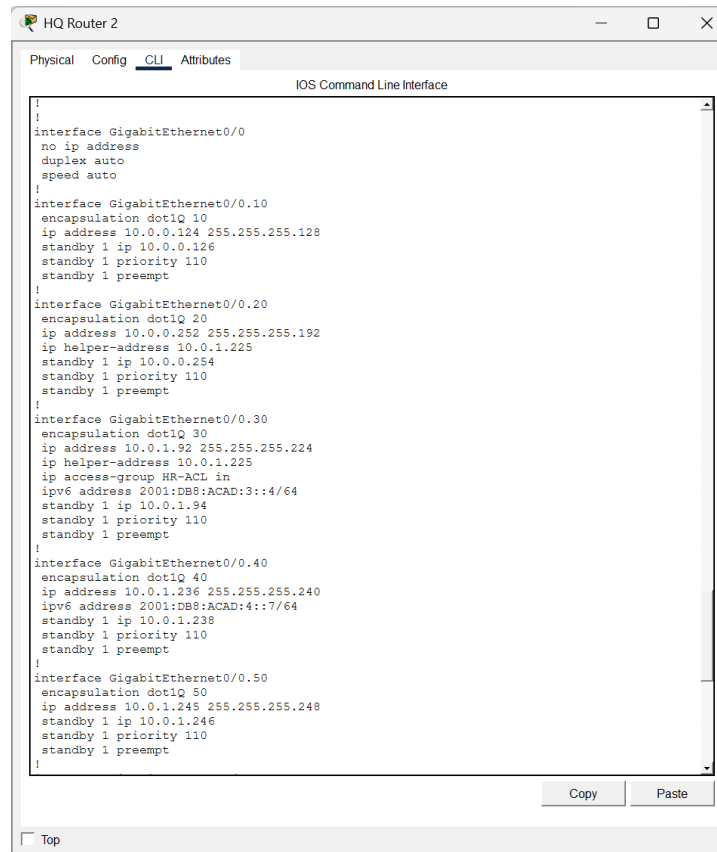
HSRP

- Set up HSRP and FHRP on HQ-Router-1 and HQ-Router-2
- WHY? In case one router goes down it can use the standby router to still communicate.
Helps with security
- HOW?
- Router(config-if)#standby 1 ip <ip-address>
- Router(config-if)#standby 1 priority <priority-number>
- Router(config-if)#standby 1 preempt



```
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 10.0.0.125 255.255.255.128
standby 1 ip 10.0.0.126
standby 1 priority 105
standby 1 preempt
standby preempt
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 10.0.0.253 255.255.255.192
ip helper-address 10.0.1.225
standby 1 ip 10.0.0.254
standby 1 priority 105
standby 1 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 10.0.1.93 255.255.255.224
ip helper-address 10.0.1.225
ip access-group HR-ACL in
ipv6 address 2001:DB8:ACAD:3::3/64
standby 1 ip 10.0.1.94
standby 1 priority 105
standby 1 preempt
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.0.1.237 255.255.255.240
ipv6 address 2001:DB8:ACAD:4::6/64
standby 1 ip 10.0.1.238
standby 1 priority 105
standby 1 preempt
!
interface GigabitEthernet0/0.50
encapsulation dot1Q 50
ip address 10.0.1.244 255.255.255.248
standby 1 ip 10.0.1.246
standby 1 priority 105
standby 1 preempt
!
```

Fig 12: HQ-Router-1 HSRP



The screenshot shows the CLI of HQ Router 2 with the following configuration:

```
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 10.0.0.124 255.255.255.128
standby 1 ip 10.0.0.126
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 10.0.0.252 255.255.255.192
ip helper-address 10.0.1.225
standby 1 ip 10.0.0.254
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 10.0.1.92 255.255.255.224
ip helper-address 10.0.1.225
ip access-group HR-ACL in
ipv6 address 2001:DB8:ACAD:3::4/64
standby 1 ip 10.0.1.94
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.0.1.236 255.255.255.240
ipv6 address 2001:DB8:ACAD:4::7/64
standby 1 ip 10.0.1.238
standby 1 priority 110
standby 1 preempt
!
interface GigabitEthernet0/0.50
encapsulation dot1Q 50
ip address 10.0.1.245 255.255.255.248
standby 1 ip 10.0.1.246
standby 1 priority 110
standby 1 preempt
!
```

Buttons for Copy and Paste are visible at the bottom right of the CLI window.

Fig 13: HQ-Router-2 HSRP

```
Enter configuration commands, one per line. End with CTRL-Z.  
R2(config)#do show standby br  
P indicates configured to preempt.  
|  
Interface  Grp  Pri  P State    Active      Standby      Virtual IP  
Gig          1   110  P Active   local       10.0.0.125   10.0.0.126  
Gig          1   110  P Active   local       10.0.0.253   10.0.0.254  
Gig          1   110  P Active   local       10.0.1.93    10.0.1.94  
Gig          1   110  P Active   local       10.0.1.237   10.0.1.238  
Gig          1   110  P Active   local       10.0.1.244   10.0.1.246  
R2(config)#
```

Fig 14: HQ-Router-2 standby showing it's the active router, the address of the standby router, and the virtual IP

3.5. Routing Configuration

OSPF

- Uses OSPF to find routing for each network
- WHY? Easier to configured instead of writing every single route
- HOW?
- Router(config)# router ospf 1
- Router(config-router)#network <network address> <wildcard mask> area 0

IPv6 static routing

- Use ipv6 route to create routes between ipv6 subnets
- WHY? So that HR and Server can communicate with Sales and Marketing
- HOW?
- Router(config)#ipv6 route <ipv6-network-address with subnet> <ipv6-address it will hop through>

```
router ospf 1
 log-adjacency-changes
 network 198.51.100.0 0.0.0.3 area 0
 network 198.51.100.4 0.0.0.3 area 0
 network 198.51.100.8 0.0.0.3 area 0
 network 198.51.100.12 0.0.0.3 area 0
 network 0.0.0.0 255.255.255.255 area 0
!
ip classless
!
ip flow-export version 9
!
ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:5::1
ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:6::1
ipv6 route 2001:DB8:ACAD:4::/64 2001:DB8:ACAD:7::1
ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:7::1
.
```

Fig 15: R5 static IPv6 routes and OSPF

```
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.0.0.127 area 0
 network 10.0.0.192 0.0.0.63 area 0
 network 10.0.1.64 0.0.0.31 area 0
 network 10.0.1.224 0.0.0.15 area 0
 network 198.51.100.8 0.0.0.3 area 0
 network 10.0.1.240 0.0.0.7 area 0
!
ip classless
!
ip flow-export version 9
!
ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:7::2
ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:7::2
.
```

Fig 16: HQ-Router-2 static IPv6 routes and OSPF

```
router ospf 1
 log-adjacency-changes
 network 198.51.100.0 0.0.0.3 area 0
 network 10.0.0.128 0.0.0.63 area 0
 network 10.0.1.128 0.0.0.31 area 0
 network 10.0.1.192 0.0.0.31 area 0
 network 0.0.0.0 255.255.255.255 area 0
!
ip classless
!
ip flow-export version 9
!
ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:5::2
ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:5::2
ipv6 route 2001:DB8:ACAD:4::/64 2001:DB8:ACAD:5::2
.
```

Fig 17: R4 static IPv6 routes and OSPF

3.6. Other Server Configuration

HTTP/HTTPS

- Set static IP address, Subnet, Default Gateway, and DNS Server
- WHY? Make it accessible by IP address and so DNS can make a domain from the IP
- HOW?

FastEthernet0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0090.0CAE.740E
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	10.0.1.229
Subnet Mask	255.255.255.240
IPv6 Configuration	
☐ Automatic	
☒ Static	
IPv6 Address	2001:DB8:ACAD:4::5 /64
Link Local Address	FE80::290:CFF:FEAE:740E

Fig 17: Configuration of HTTP/HTTPS IP address, Subnet

Global Settings

Display Name HTTP/HTTPS

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway 10.0.1.238

DNS Server 10.0.1.226

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway 2001:DB8:ACAD:4::7

DNS Server

Device Clock: 12:43:24 Sat Mar 7 2026 UTC

PTP

Profile Power

☐ Enable ☒ Disable

Fig 18: Configuration of HTTP/HTTPS Default Gateway, DNS

- Enable HTTP/HTTPS and set up index.html
- WHY? Let the PCs access index.html via the IP address
- HOW?

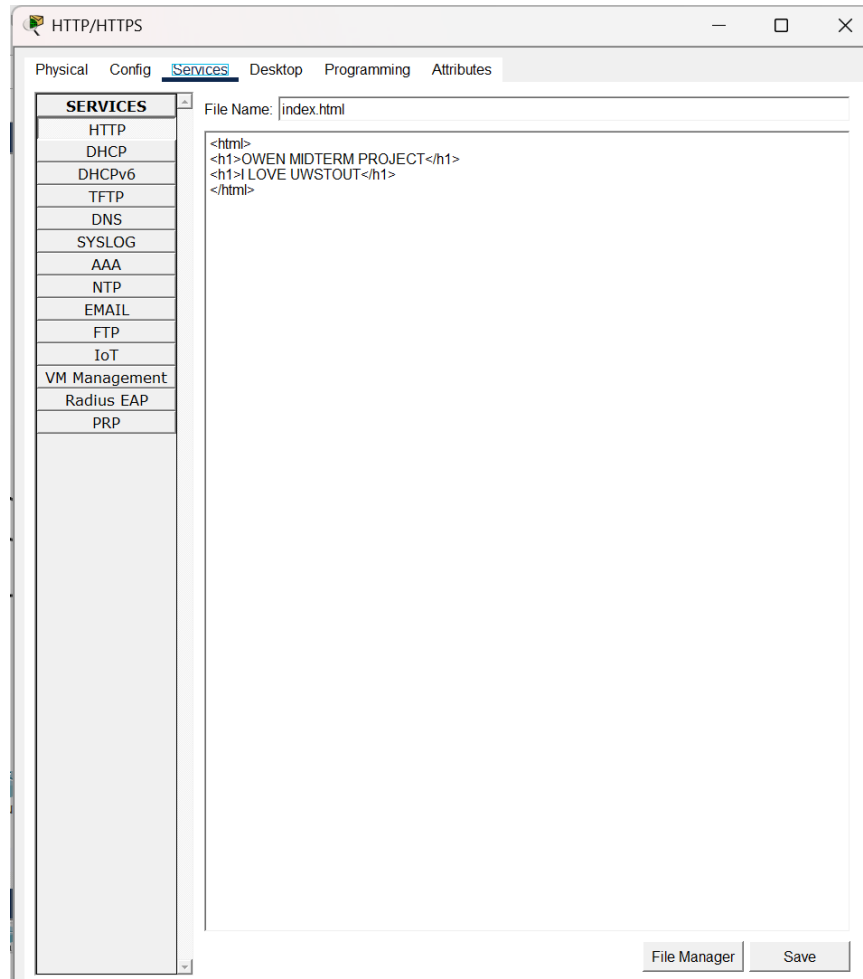


Fig 19: Configuration of the index.html file

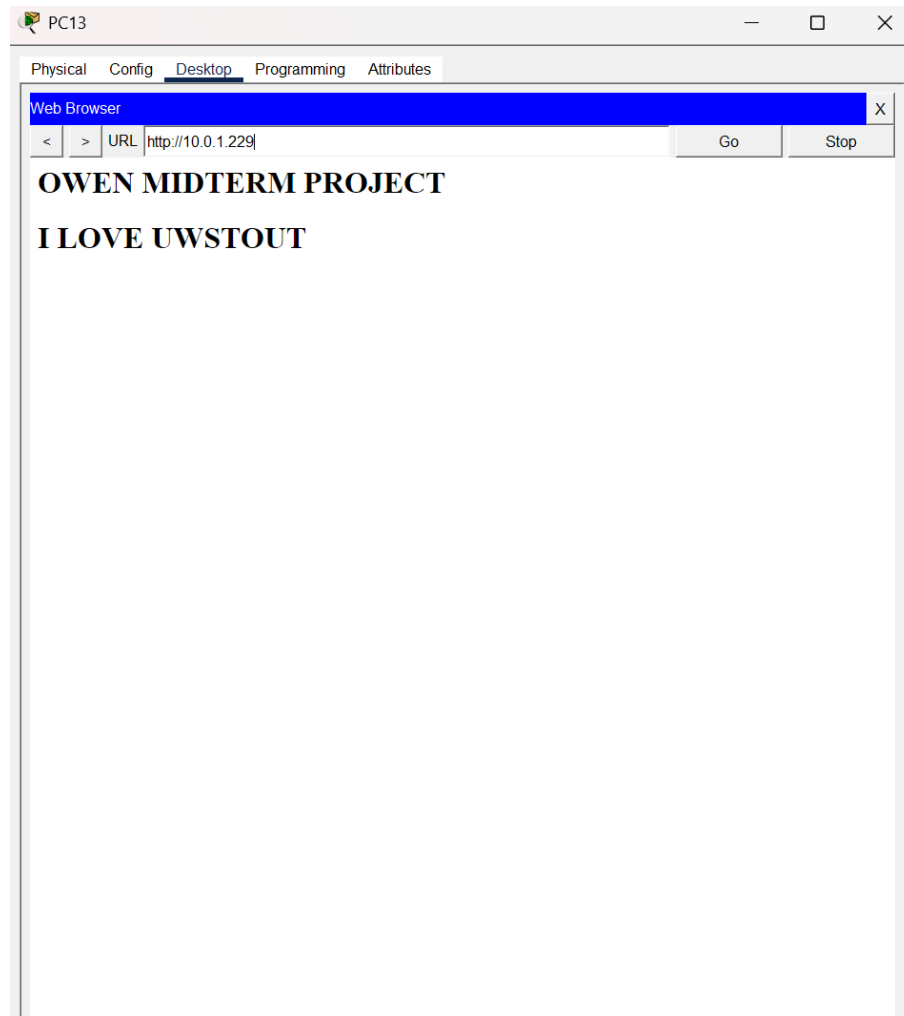
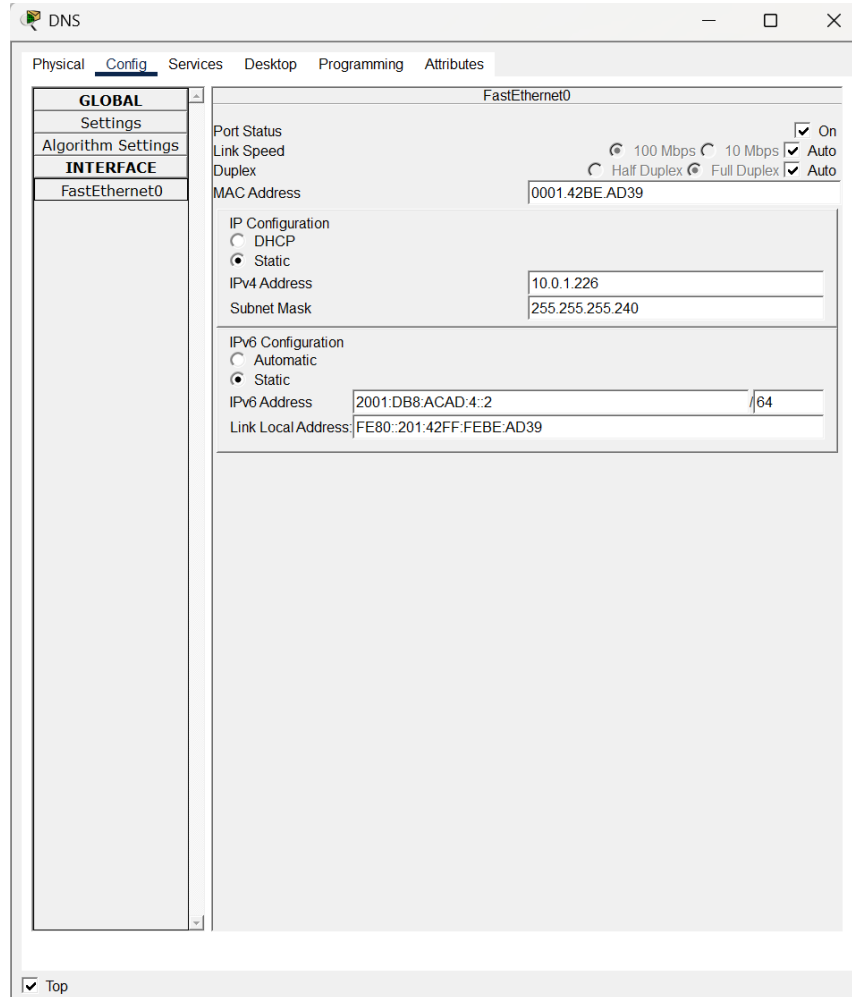


Fig 20: PC13 in IT accessing the HTTP/HTTPS server with the IP address

DNS

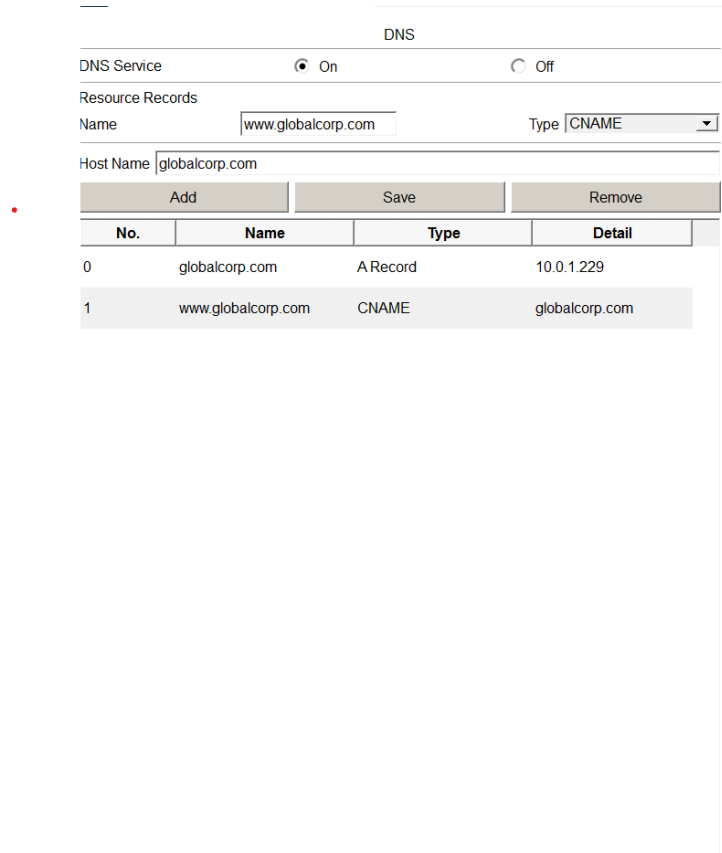
- Set static IP address, Subnet, Default Gateway, and DNS Server
- WHY? So other PCs can configure the DNS Server in their IP configurations
- HOW?



The screenshot shows a network configuration window titled "DNS". The "Config" tab is selected, and the "FastEthernet0" interface is chosen. The configuration is divided into "GLOBAL" and "INTERFACE" sections. Under "INTERFACE", the "FastEthernet0" settings are displayed. The "Port Status" is "On". The "Link Speed" is set to "100 Mbps" and "Auto". The "Duplex" is set to "Full Duplex" and "Auto". The "MAC Address" is "0001.42BE.AD39". The "IP Configuration" is set to "Static", with the "IPv4 Address" set to "10.0.1.226" and the "Subnet Mask" set to "255.255.255.240". The "IPv6 Configuration" is also set to "Static", with the "IPv6 Address" set to "2001:DB8:ACAD:4::2" and the "Link Local Address" set to "FE80::201:42FF:FEBE:AD39".

Fig 21: Configuration for the DNS Server IP Address, Subnet

- Enable DNS and set up 2 records
- 1. globalcorp.com
 - Type: A Record
 - Address: 10.0.1.229
- 2. www.globalcorp.com
 - Type: CNAME
 - Host Name: globalcorp.com
- WHY? Let's the PCs connect to HTTP/HTTPS server via a domain name
- HOW?



The screenshot shows a DNS configuration interface. At the top, there's a section for "DNS" with a "DNS Service" toggle set to "On". Below this, there's a "Resource Records" section. It includes a "Name" field with "www.globalcorp.com", a "Type" dropdown set to "CNAME", and a "Host Name" field with "globalcorp.com". There are "Add", "Save", and "Remove" buttons. Below these fields is a table listing the configured records.

No.	Name	Type	Detail
0	globalcorp.com	A Record	10.0.1.229
1	www.globalcorp.com	CNAME	globalcorp.com

Fig 22: DNS configuration for the two domain names

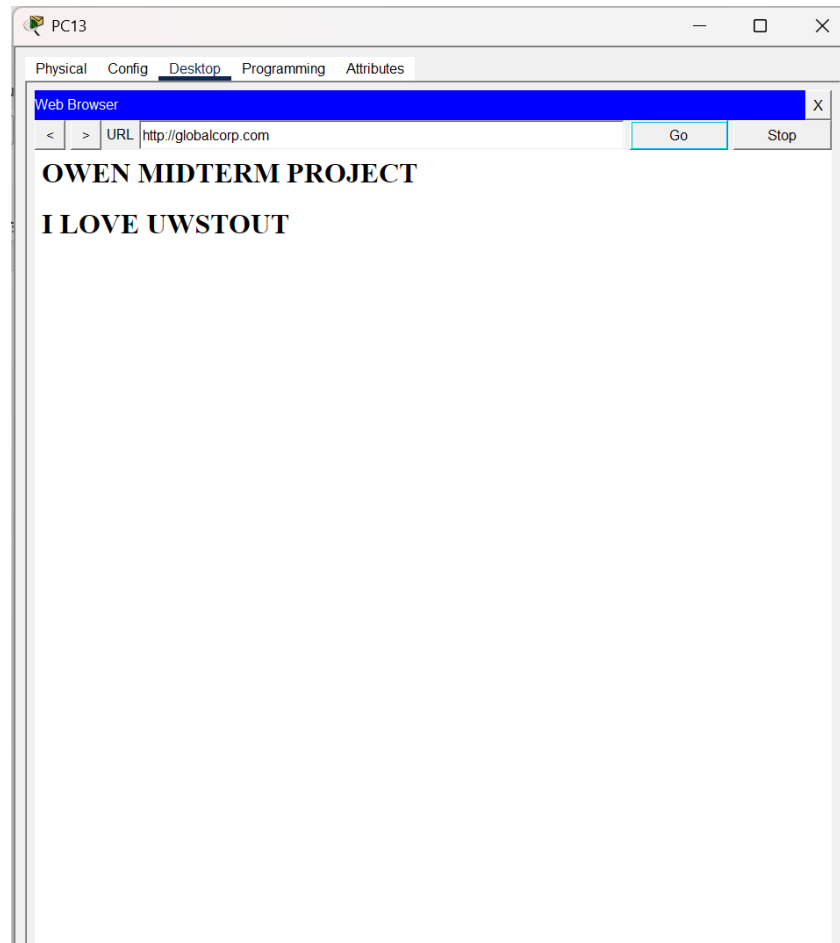


Fig 23: PC13 accessing HTTP/HTTPS via globalcorp.com

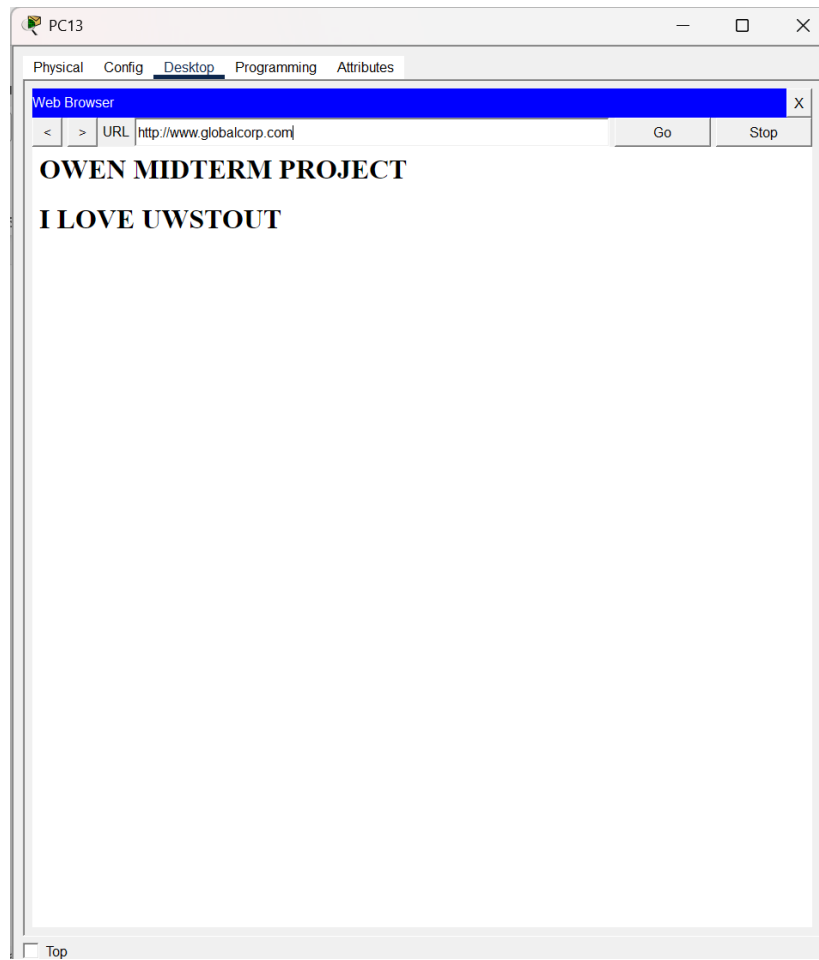


Fig 24: PC13 accessing HTTP/HTTPS via www.globalcorp.com

FTP

- Set static IP address, Subnet, Default Gateway, and DNS Server
- WHY? So other PCs can get into FTP with the address
- HOW?

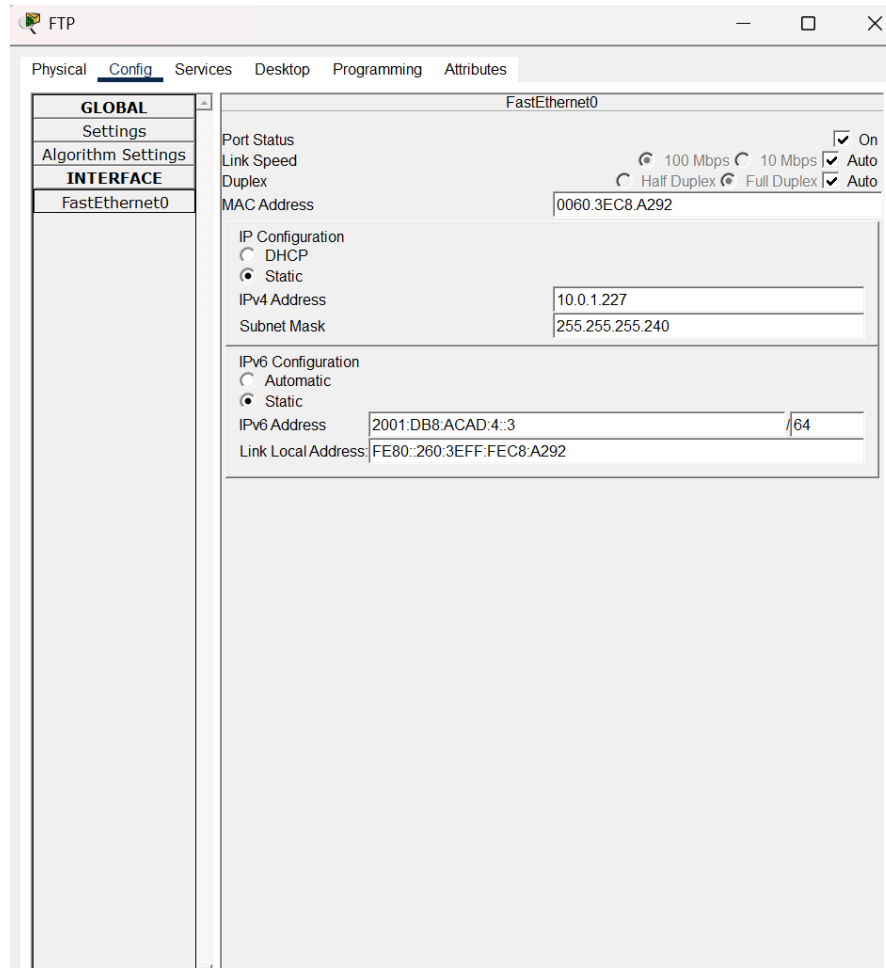
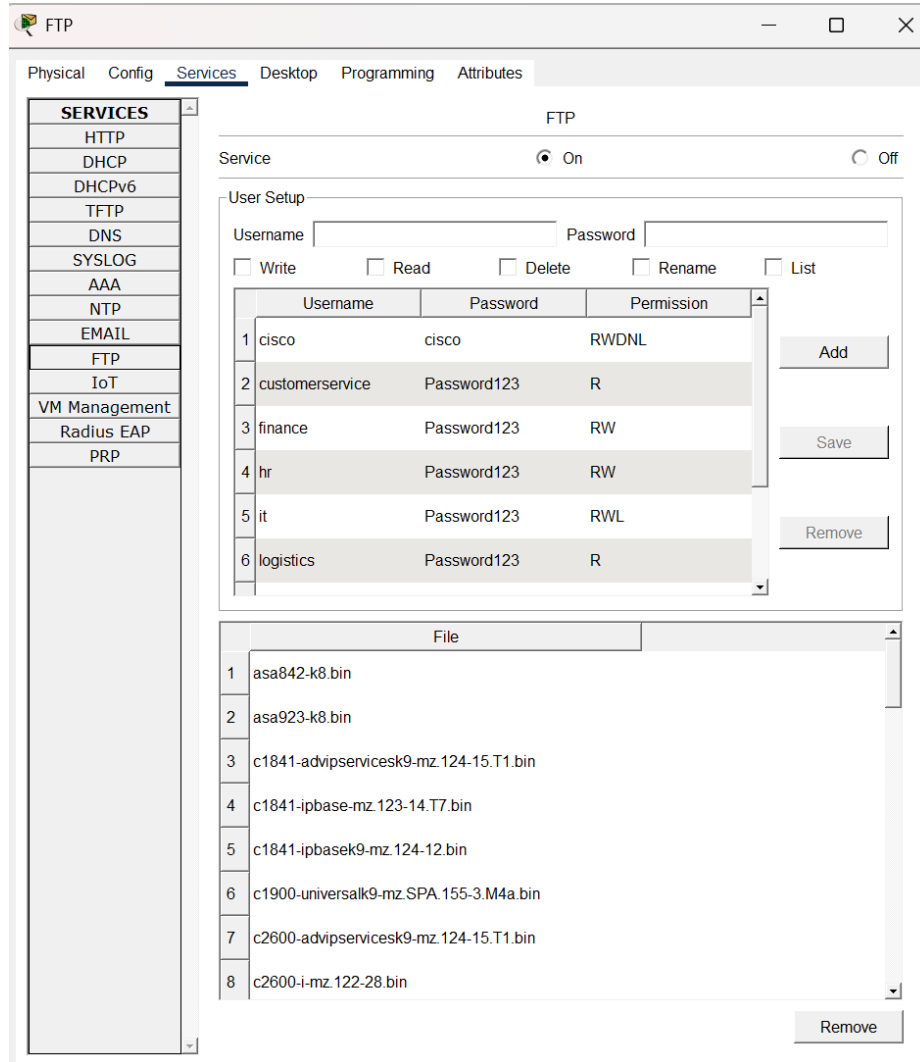


Fig 25: FTP Server configuration

- Set profiles for each department which includes usernames, passwords, and different restrictions
- WHY? Different departments have different restrictions within the FTP on what they can do
- HOW?

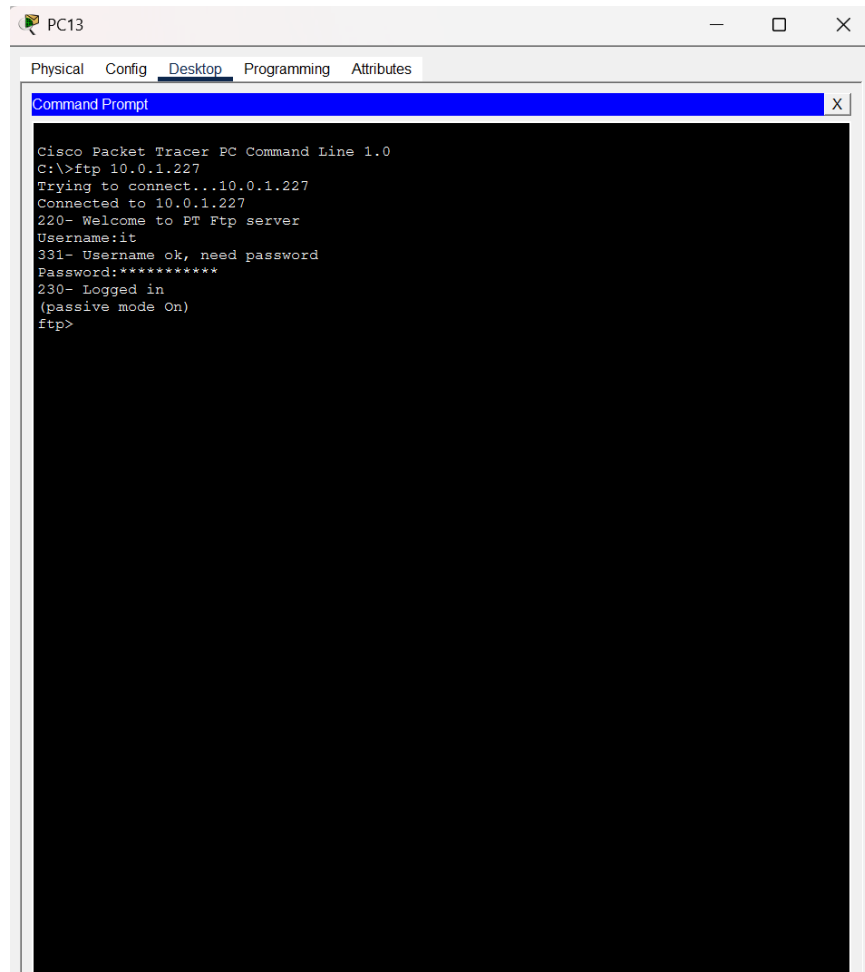


The screenshot shows the 'Services' tab in a configuration window. The 'FTP' service is set to 'On'. Under 'User Setup', there is a table of users with columns for Username, Password, and Permission. Below this is a 'File' list showing various binary files.

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	customerservice	Password123	R
3	finance	Password123	RW
4	hr	Password123	RW
5	it	Password123	RWL
6	logistics	Password123	R

	File
1	asa842-k8.bin
2	asa923-k8 bin
3	c1841-advipservicesk9-mz.124-15.T1.bin
4	c1841-ipbase-mz.123-14.T7.bin
5	c1841-ipbasek9-mz.124-12.bin
6	c1900-universalk9-mz.SPA.155-3.M4a.bin
7	c2600-advipservicesk9-mz.124-15.T1.bin
8	c2600-i-mz.122-28.bin

Fig 26: list of FTP accounts and different permissions



The screenshot shows a Cisco Packet Tracer interface with a window titled 'PC13'. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 10.0.1.227
Trying to connect...10.0.1.227
Connected to 10.0.1.227
220- Welcome to FT Ftp server
Username:it
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
```

Fig 27: PC13 in IT accessing the FTP via Command Prompt

3.7. Access Control Lists

- ACL put on routers to deny or permit certain Ips
- WHY? Different departments have different access to the servers, and the routers have to deny or permit the traffic
- HOW?
- Router(config)#ip access-list extended <acl-name>
- Router(config-extd-acl)# <order-number> <permit or deny>
<protocol> <surce-ip> <wildcard-mask> <destination-ip> <wildcard-mask> <port options>
- Router(config-extd-acl)#exit
- Router(config)#int <interface-name>
- Router(config-if)#ip access-group <acl-name> <in/out>

```
R2#show access-list
Extended IP access list HR-ACL
 10 deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq www
 20 deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq 443
 30 permit ip any any (1305 match(es))

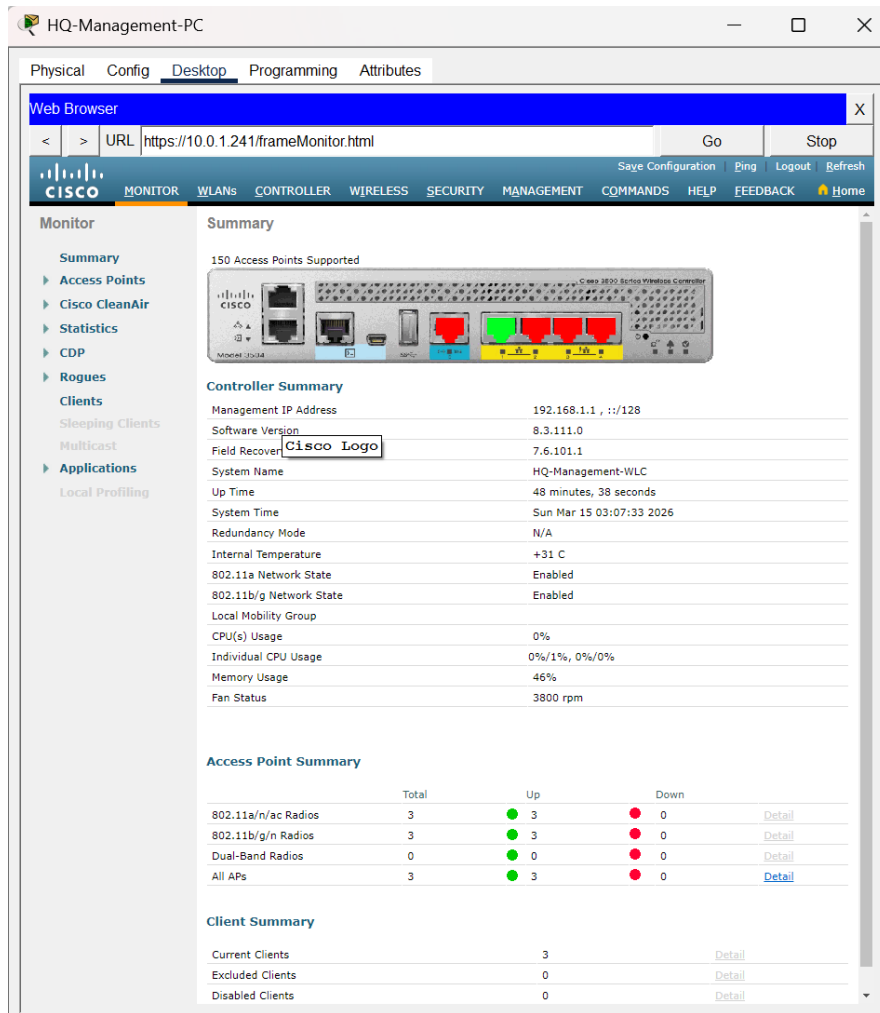
R2#|
```

Fig 28: Example of HQ-Router-2 with it's ACL to block HTTP/HTTPS for HR

3.8. Wireless Network and RADIUS Setup

WLC

- Set up WLC as 10.0.1.241
- WHY? Have the HQ-Management-PC access the WLC
- HOW?



Web Browser

URL: <https://10.0.1.241/frameMonitor.html>

Go Stop

Save Configuration Ping Logout Refresh

CISCO MONITOR WLANS CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK Home

Monitor

Summary

150 Access Points Supported

Controller Summary

Management IP Address	192.168.1.1, ::1/128
Software Version	8.3.111.0
Field Recover	7.6.101.1
System Name	HQ-Management-WLC
Up Time	48 minutes, 38 seconds
System Time	Sun Mar 15 03:07:33 2026
Redundancy Mode	N/A
Internal Temperature	+31 C
802.11a Network State	Enabled
802.11b/g Network State	Enabled
Local Mobility Group	
CPU(s) Usage	0%
Individual CPU Usage	0%/1%, 0%/0%
Memory Usage	46%
Fan Status	3800 rpm

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	3	3	0	Detail
802.11b/g/n Radios	3	3	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	3	3	0	Detail

Client Summary

Current Clients	3	Detail
Excluded Clients	0	Detail
Disabled Clients	0	Detail

Fig 29: Website for WLC and shows that a port is active on it

SSIDs.

- Configure SSIDs on the WLC website
- WHY? Set specific Wi-Fis for the separate departments
- HOW?



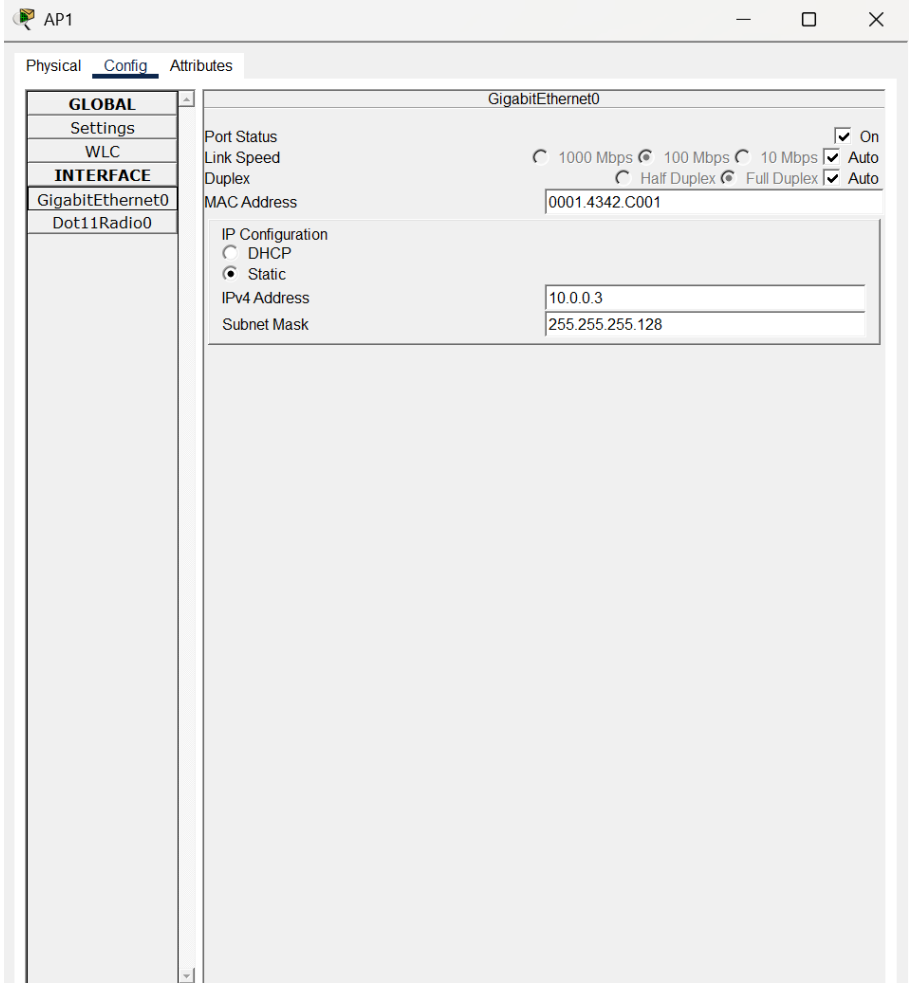
The screenshot shows the Cisco WLC web interface. The top navigation bar includes links for MONITOR, WLANs (highlighted), CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, COMMANDS, HELP, and FEEDBACK. On the left, a sidebar shows 'WLANs' and 'Advanced' options. The main content area is titled 'WLANs' and contains a table of configured WLANs. Above the table, there are links for '[Change Filter]' and '[Clear Filter]', and a 'Create New' button with a dropdown arrow and a 'Go' button.

☐	WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies	
☐	1	WLAN	HQ-Management	HQ-Management	Enabled	[WPA2][Auth(PSK)]	Remove
☐	2	WLAN	IT-WiFi	IT	Enabled	[WPA2][Auth(PSK)]	Remove
☐	3	WLAN	Finance-WiFi	Finance	Enabled	[WPA2][Auth(PSK)]	Remove
☐	4	WLAN	HR-WiFi	HR	Enabled	[WPA2][Auth(PSK)]	Remove

Fig 30: List of SSIDs configured on the WLC for the HQ departments

Lightweight APs.

- Configure IP information and WLC Address
- WHY? Needs a WLC Address to know how to get to the WLC
- HOW?



The screenshot shows the configuration window for AP1. The 'Config' tab is active, and the 'GigabitEthernet0' interface is selected. The configuration includes the following settings:

Category	Setting	Value
Physical	Port Status	☒ On
	Link Speed	☒ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
	Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
	MAC Address	0001.4342.C001
IP Configuration	IP Configuration	☐ DHCP ☒ Static
	IPv4 Address	10.0.0.3
	Subnet Mask	255.255.255.128

Fig 31: Static IP information for AP1

- Get IP information and WLC address from DHCP
- WHY? Finance and HR have DHCP for their IP information
- HOW?

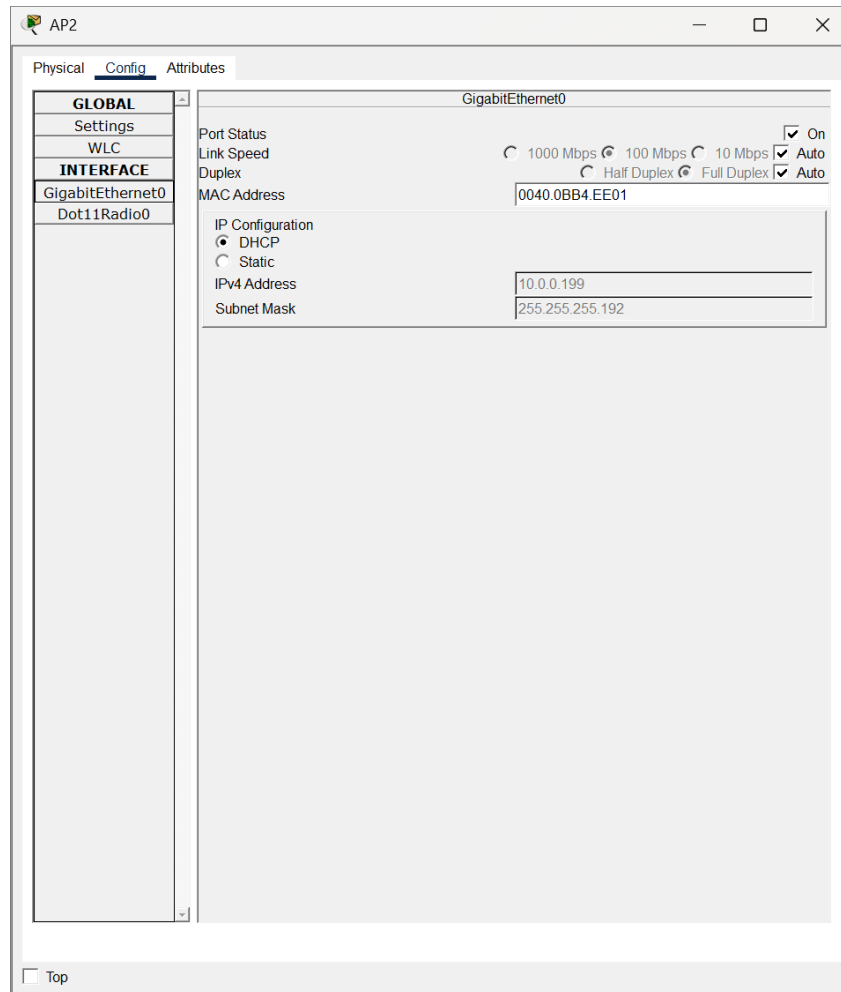


Fig 32: DHCP IP information for AP2

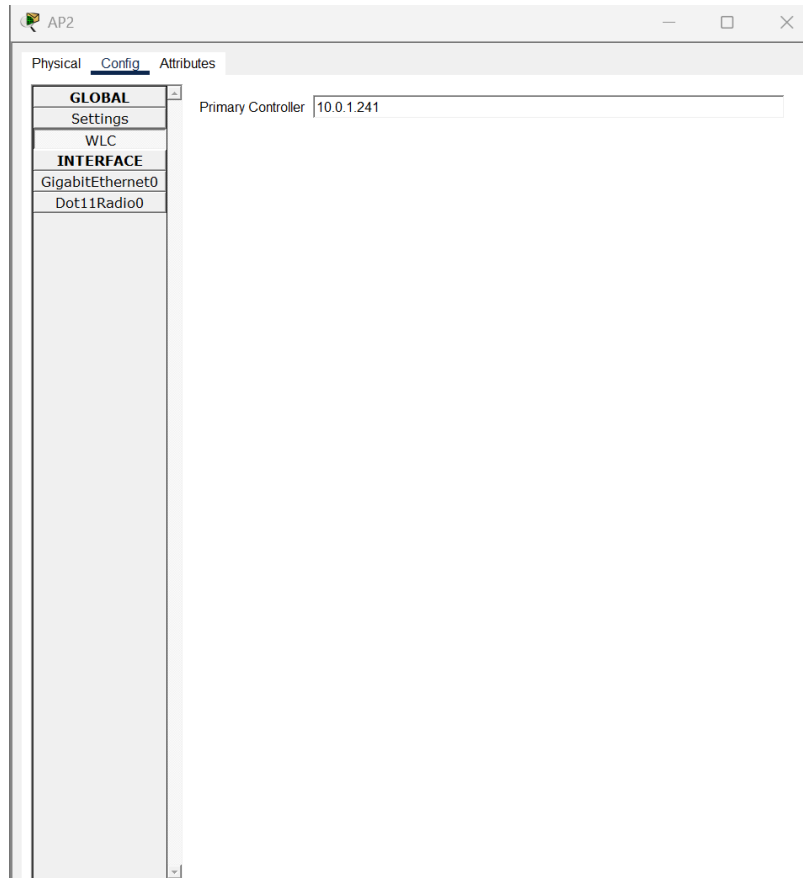
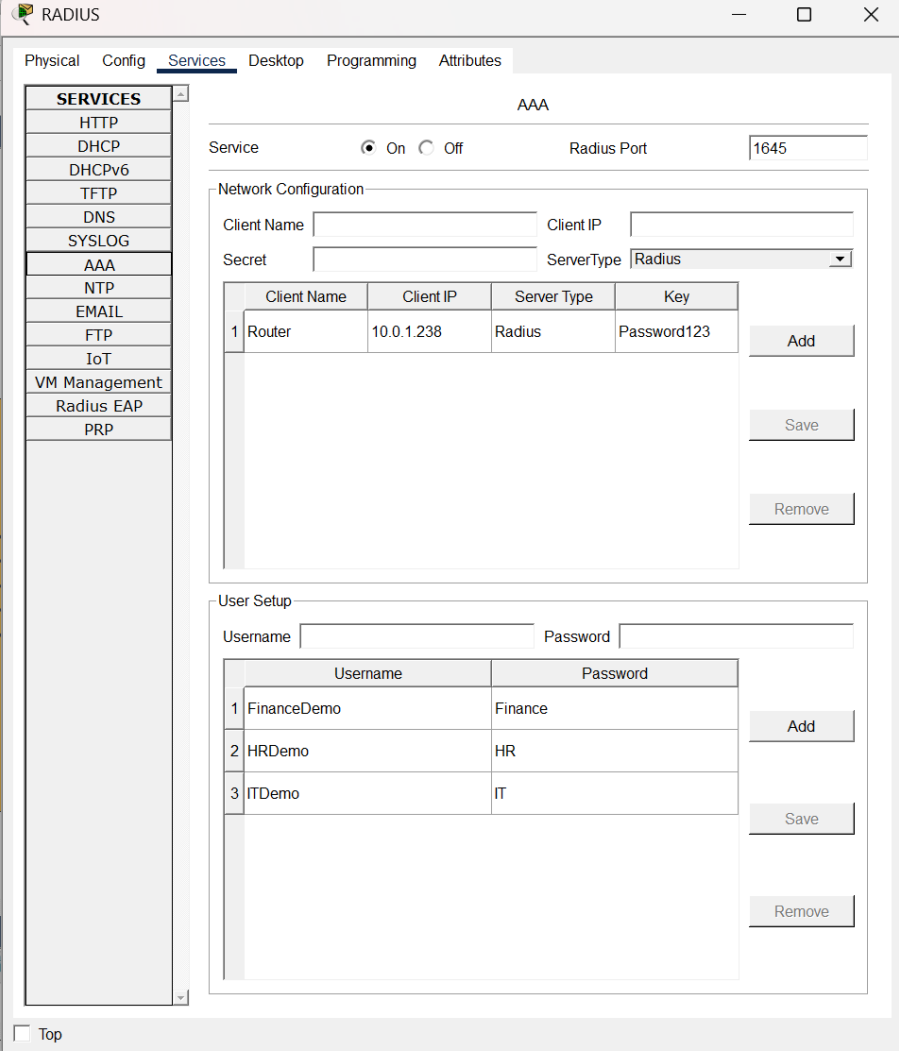


Fig 33: WLC Address configured on AP2

RADIUS

- Configure AAA with the Router and set up users
- WHY? Adds security for each department when connecting to wireless
- HOW?



The screenshot shows the RADIUS configuration window with the 'Services' tab selected. The 'AAA' service is configured with 'On' selected for the Service and '1645' for the Radius Port. The 'Network Configuration' section includes fields for Client Name, Client IP, Secret, and Server Type (set to Radius). Below this is a table for adding network configurations.

	Client Name	Client IP	Server Type	Key	
1	Router	10.0.1.238	Radius	Password123	Add

Buttons for 'Save' and 'Remove' are located to the right of the table.

The 'User Setup' section includes fields for Username and Password. Below this is a table for adding users.

	Username	Password	
1	FinanceDemo	Finance	Add
2	HRDemo	HR	
3	ITDemo	IT	

Buttons for 'Save' and 'Remove' are located to the right of the table.

At the bottom left, there is a 'Top' button.

Fig 34: Radius server with AAA set up for each department

4. Testing and Verification

- Ping Tests:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=10ms TTL=127
Reply from 10.0.0.1: bytes=32 time<1ms TTL=127
Reply from 10.0.0.1: bytes=32 time=2ms TTL=127
Reply from 10.0.0.1: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 10.0.0.198

Pinging 10.0.0.198 with 32 bytes of data:

Reply from 10.0.0.198: bytes=32 time<1ms TTL=127
Reply from 10.0.0.198: bytes=32 time<1ms TTL=127
Reply from 10.0.0.198: bytes=32 time=1ms TTL=127
Reply from 10.0.0.198: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.0.198:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.0.1.225

Pinging 10.0.1.225 with 32 bytes of data:

Reply from 10.0.1.225: bytes=32 time<1ms TTL=127
Reply from 10.0.1.225: bytes=32 time=15ms TTL=127
Reply from 10.0.1.225: bytes=32 time<1ms TTL=127
Reply from 10.0.1.225: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.1.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 3ms
```

Fig 35: HR PC pinging a device from IT, Finance, Servers

```
C:\>ping 10.0.1.242

Pinging 10.0.1.242 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.242: bytes=32 time<1ms TTL=127
Reply from 10.0.1.242: bytes=32 time<1ms TTL=127
Reply from 10.0.1.242: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.1.242:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.1.3

Pinging 10.0.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.0.1.3: bytes=32 time=4ms TTL=125
Reply from 10.0.1.3: bytes=32 time=4ms TTL=125

Ping statistics for 10.0.1.3:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>ping 10.0.1.102

Pinging 10.0.1.102 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.102: bytes=32 time=5ms TTL=125
Reply from 10.0.1.102: bytes=32 time=4ms TTL=125
Reply from 10.0.1.102: bytes=32 time=5ms TTL=125

Ping statistics for 10.0.1.102:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 5ms, Average = 4ms
```

Fig 36: HR PC pinging a device from WLC, Sales, Support

```
C:\>ping 10.0.0.129

Pinging 10.0.0.129 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.129: bytes=32 time=5ms TTL=125
Reply from 10.0.0.129: bytes=32 time=4ms TTL=125
Reply from 10.0.0.129: bytes=32 time=4ms TTL=125

Ping statistics for 10.0.0.129:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 5ms, Average = 4ms

C:\>ping 10.0.1.198

Pinging 10.0.1.198 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.198: bytes=32 time=6ms TTL=125
Reply from 10.0.1.198: bytes=32 time=4ms TTL=125
Reply from 10.0.1.198: bytes=32 time=2ms TTL=125

Ping statistics for 10.0.1.198:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 4ms

C:\>ping 10.0.1.133

Pinging 10.0.1.133 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.133: bytes=32 time=2ms TTL=125
Reply from 10.0.1.133: bytes=32 time=3ms TTL=125
Reply from 10.0.1.133: bytes=32 time=2ms TTL=125

Ping statistics for 10.0.1.133:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 10.0.1.165

Pinging 10.0.1.165 with 32 bytes of data:

Reply from 10.0.1.165: bytes=32 time=2ms TTL=125
Reply from 10.0.1.165: bytes=32 time=2ms TTL=125
Reply from 10.0.1.165: bytes=32 time=2ms TTL=125
```

Fig 37: HR PC pinging a device from Logistics, Marketing, Research, Customer Service

- Traceroute Analysis

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 10.0.0.1

Tracing route to 10.0.0.1 over a maximum of 30 hops:

  1  10 ms    0 ms    0 ms    10.0.1.92
  2   0 ms    0 ms   12 ms   10.0.0.1

Trace complete.

C:\>tracert 10.0.1.225

Tracing route to 10.0.1.225 over a maximum of 30 hops:

  1   6 ms    0 ms    0 ms    10.0.1.92
  2   0 ms    0 ms    0 ms   10.0.1.225

Trace complete.

C:\>tracert 10.0.1.3

Tracing route to 10.0.1.3 over a maximum of 30 hops:

  1   0 ms    0 ms    0 ms    10.0.1.92
  2   *      0 ms    0 ms   198.51.100.14
  3   1 ms    2 ms    2 ms   198.51.100.5
  4   1 ms    2 ms    1 ms   10.0.1.3

Trace complete.

C:\>tracert 10.0.0.129

Tracing route to 10.0.0.129 over a maximum of 30 hops:

  1   0 ms    0 ms    0 ms    10.0.1.92
  2   1 ms   10 ms    1 ms   198.51.100.10
  3   2 ms    2 ms    0 ms   198.51.100.1
  4   0 ms    0 ms    1 ms   10.0.0.129

Trace complete.

C:\>|
```

Fig 38: HR PC tracing a route to another HQ department, Servers, and both Branch Offices

- **HSRP Test:**

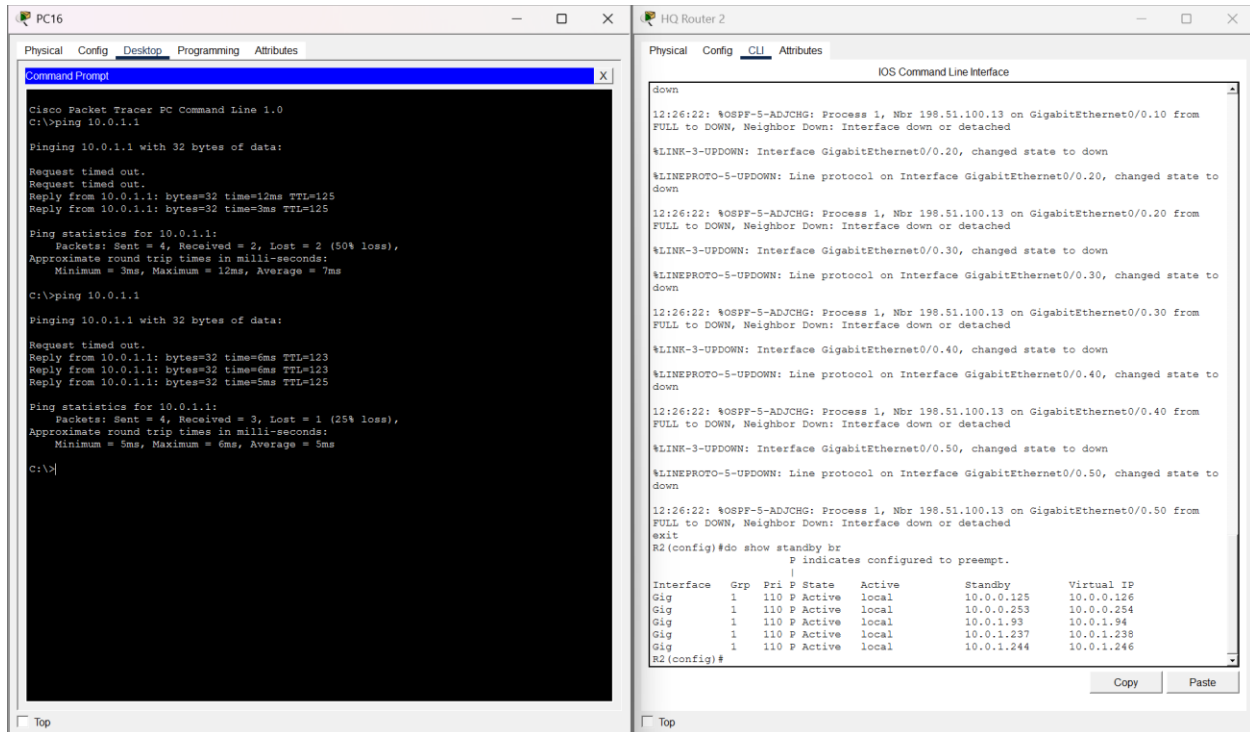


Fig 39: PC16 in Finance pinging somewhere and HQ-Router-2 being inactive with HQ-Router-1 being active now

- HTTP and DNS Tests:

HTTP/HTTPS

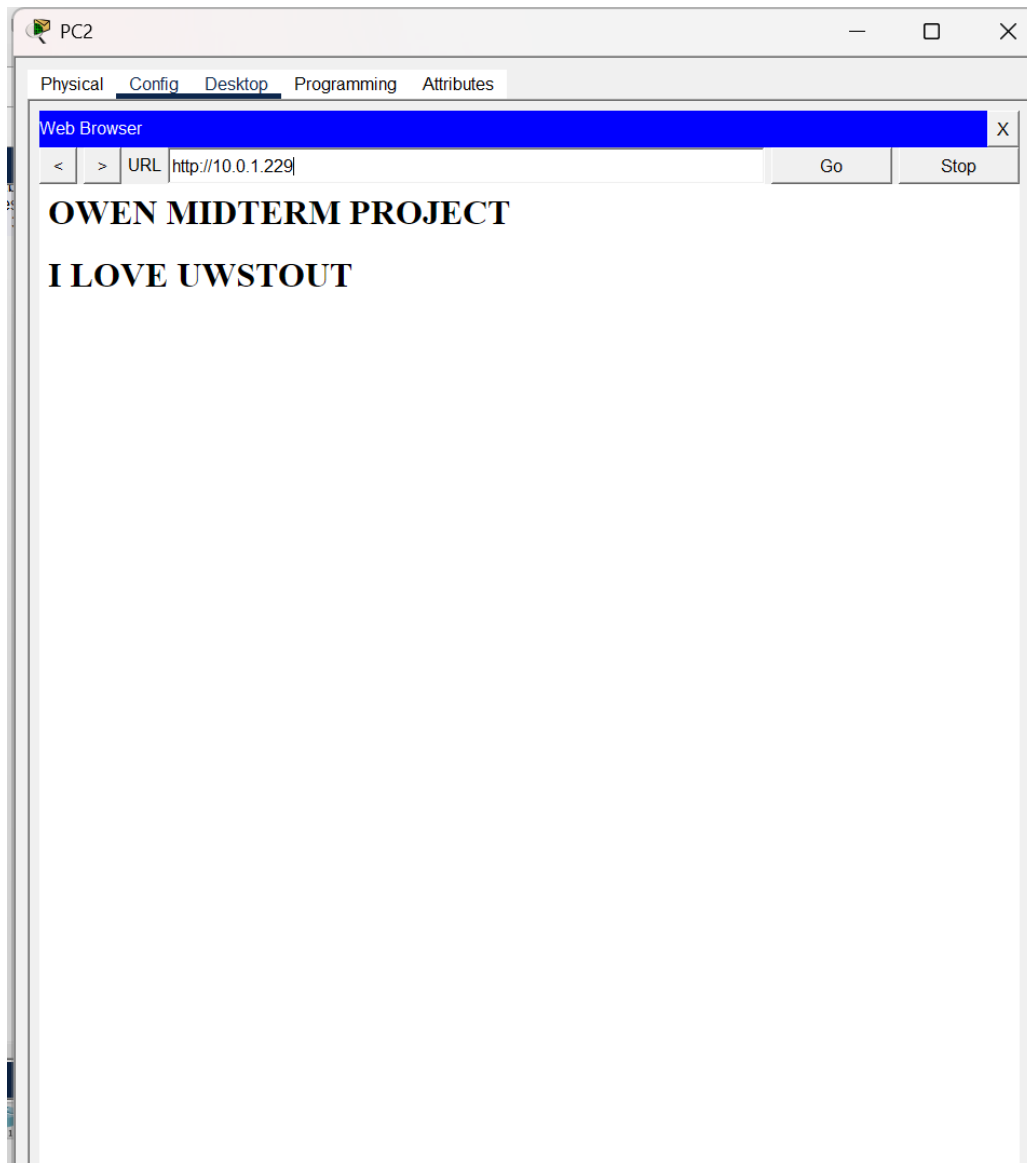


Fig 40: PC2 in Sales accessing HTTP/HTTPS with the IP address

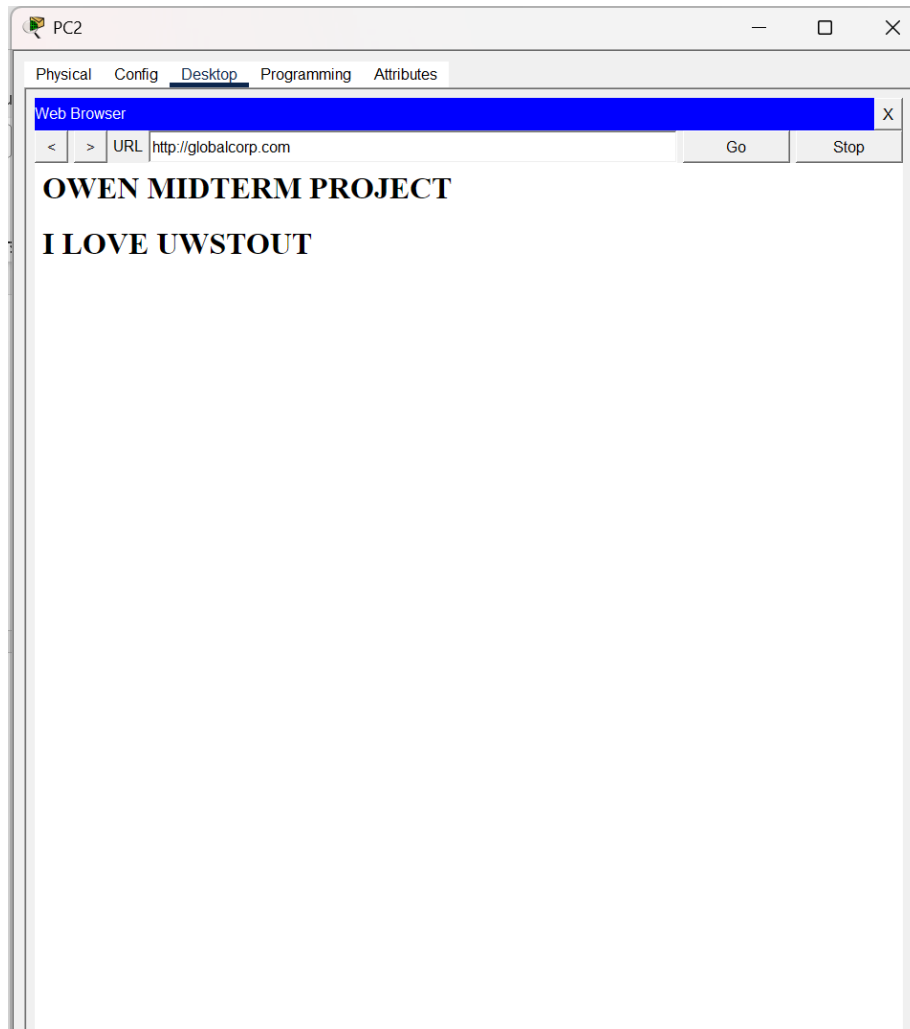


Fig 41: PC2 in Sales accessing the DNS with globalcorp.com

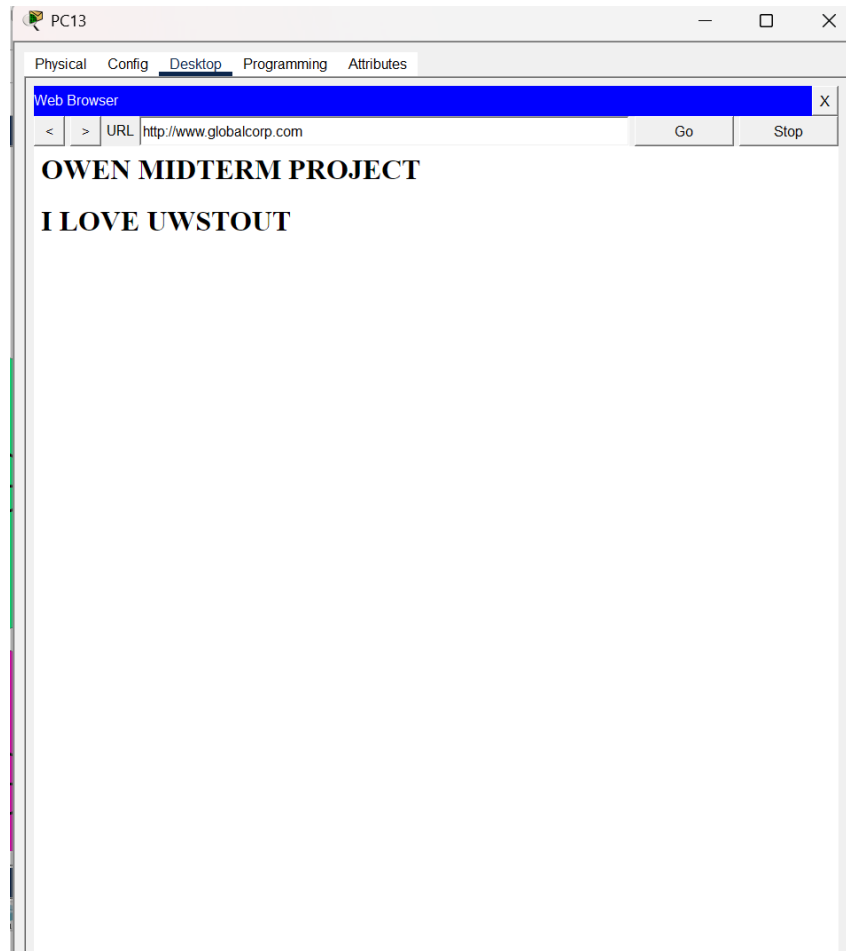
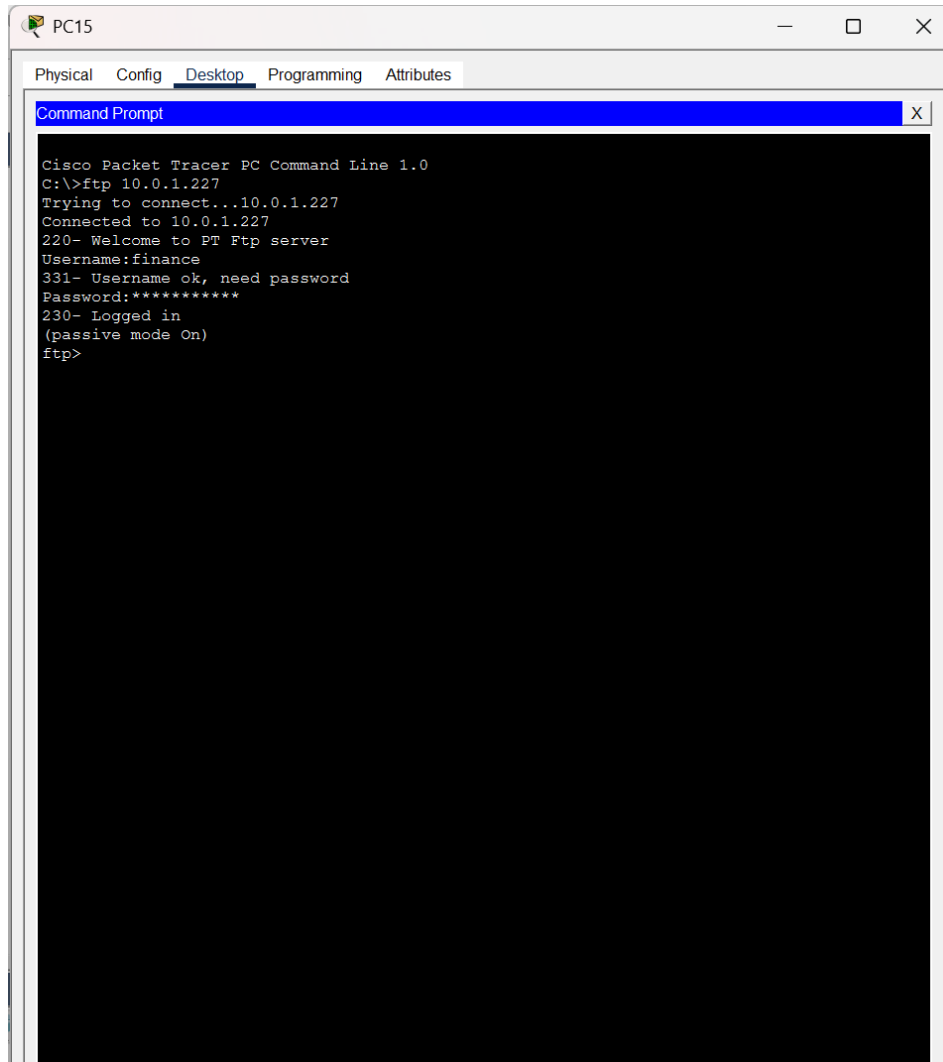


Fig 42: PC13 in IT accessing the DNS with www.globalcorp.com

- **FTP Test:**



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 10.0.1.227
Trying to connect...10.0.1.227
Connected to 10.0.1.227
220- Welcome to PT Ftp server
Username:finance
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
```

Fig 43: PC15 in Finance accessing FTP in the command prompt

- **ACL Test:** Show the effects of ACL lists

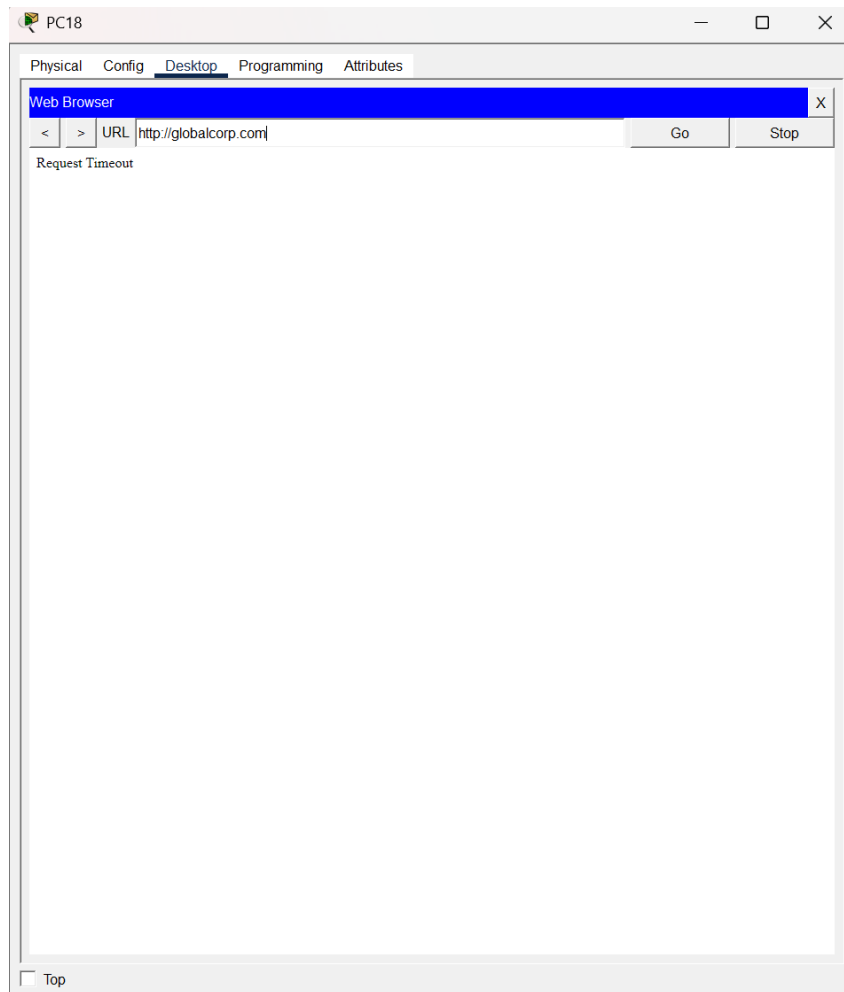


Fig 44: HR being blocked from accessing HTTP/HTTPS with an ACL

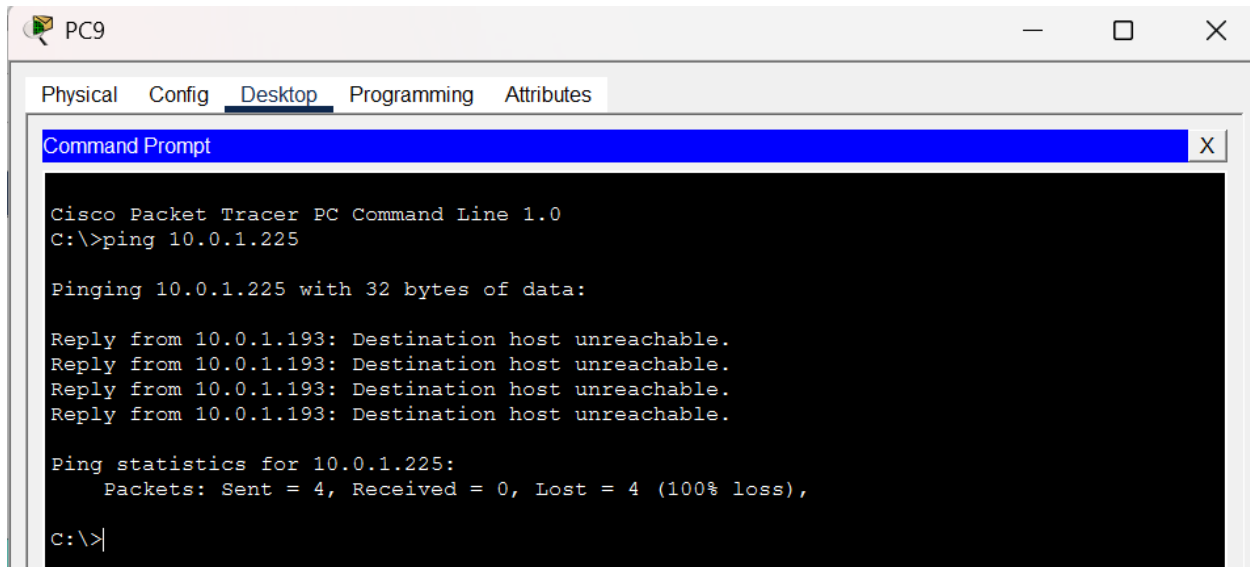


Fig 45: Research being blocked from pinging the DHCP server with an ACL

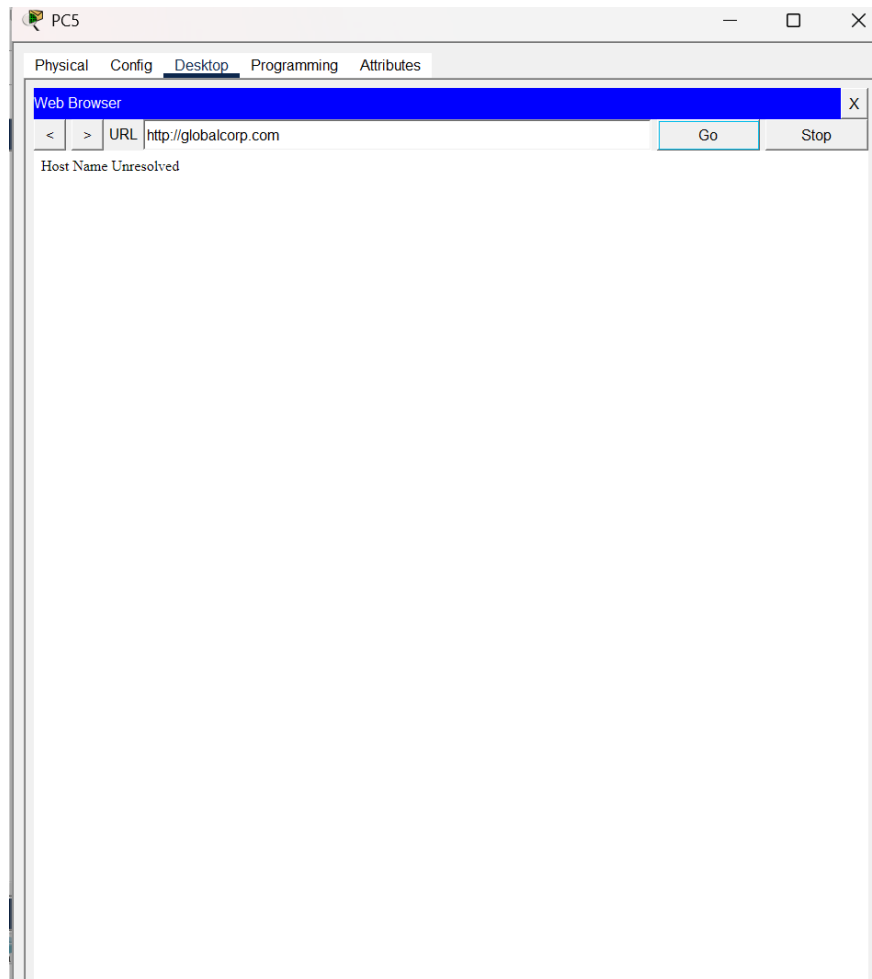


Fig 46: PC5 in Logistics being blocked from accessing the HTTP/HTTPS via DNS

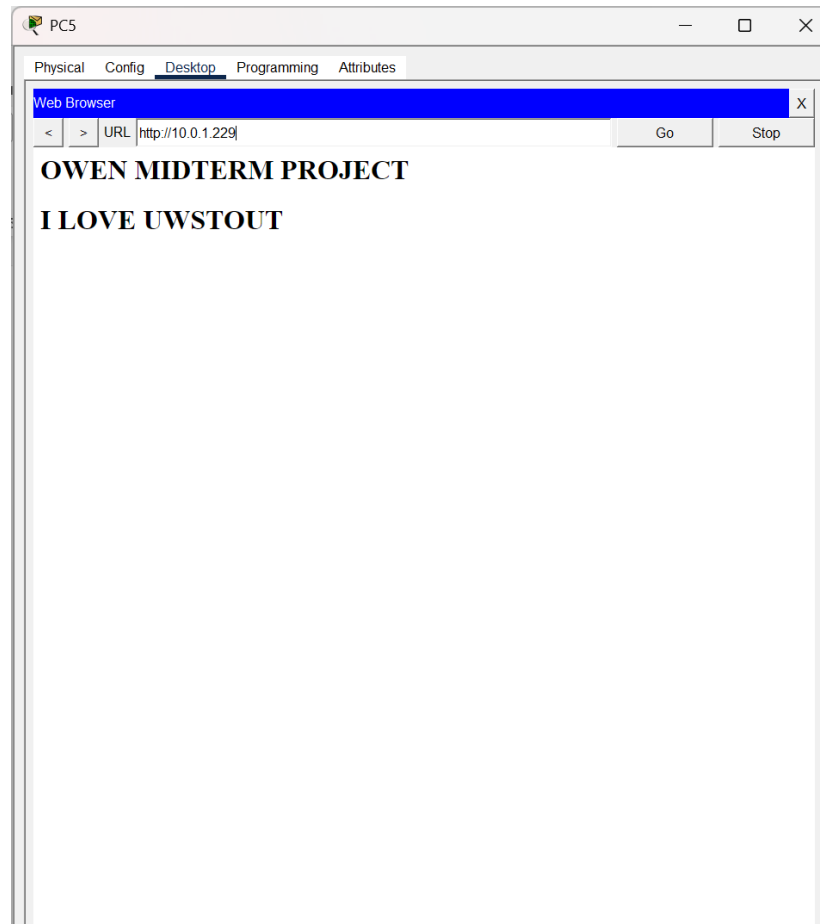


Fig 47: PC5 in Logistics being able to access HTTP/HTTPS with IP address

- **Wireless Network Tests:**

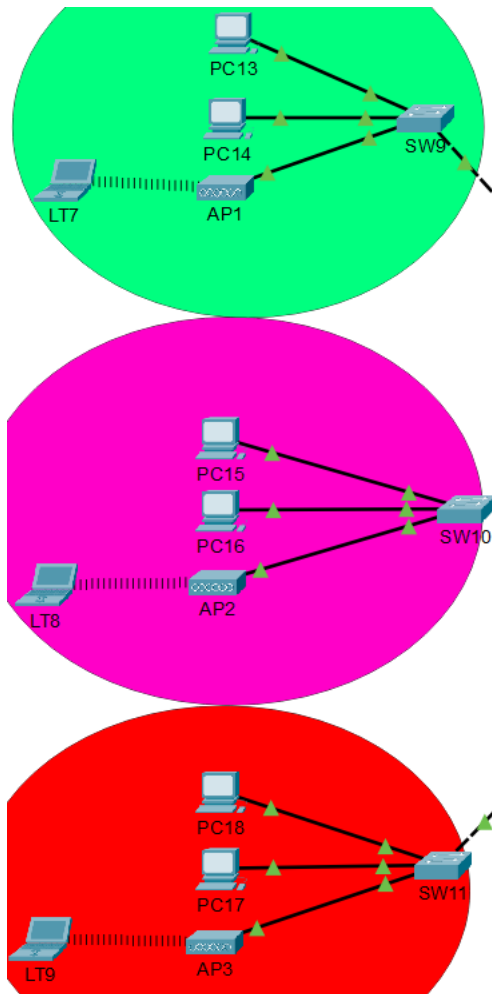


Fig 48: IT, Finance, and HR having their wireless network connected

5. Challenges and Solutions

- Issues:
 - Routing OSPF with a lot of departments
 - Configuring HTTP/HTTPS Server
 - Configuring DNS Server
 - Configuring WLC
 - Configuring RADIUS
- Provide solutions implemented to resolve them.
 - Using multiple addresses to test OSPF and get the networks connected
 - Google how to make HTTP/HTTPS server
 - Google how to make DNS server
 - Watch a video on how to configure a WLC
 - Watch a video on how to configure a RADIUS Server

6. Conclusion

- Summary of the project's success.
 - All departments can communicate with routing, can communicate between departments, can utilize the servers when needed, can connect to Wireless
- Key learnings from the project.
 - How to build an advanced network
 - How to configure servers
 - How to implement OSPF with HSRP
 - How to configure WLC
 - How to configure RADIUS